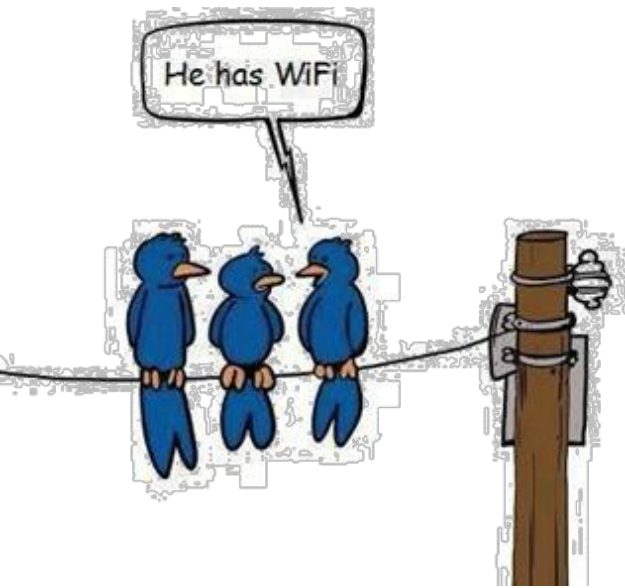


ECSE 4560/ 6560

Modern Communication Systems

Lecture 10: OFDM Synchronization



Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Synchronization**
- Projects
 - Project Topics and Template posted on Piazza
 - Group formation [MCS Fall25 project.xlsx](#)
 - Discuss project during office hours
- HW3/Lab3 posted. Due **Friday, 10/10**

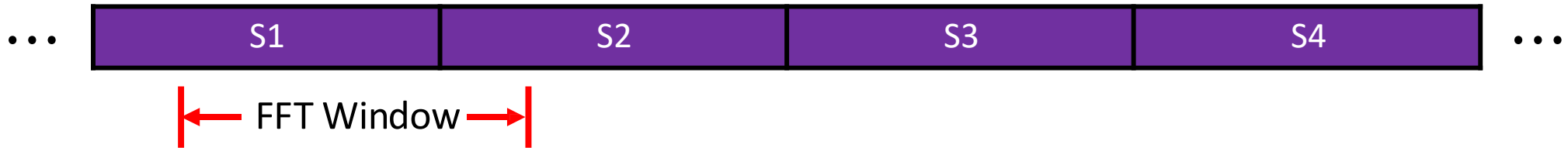
Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - Parts of Chapter 1,2,3,4,7
- An Adaptive Orthogonal Frequency Division Multiplexing Baseband Modem for Wideband Wireless Channels; by Jit Ken Tan, MIT Thesis
 - OFDM Decoding and Synchronization
 - Lab 4 Implementation

Today's Class: OFDM Synchronization

- OFDM Synchronization
 - Packet detection
 - CFO correction
 - Channel Estimation
 - SFO and Residual Phase correction
- 
- 

OFDM Symbol



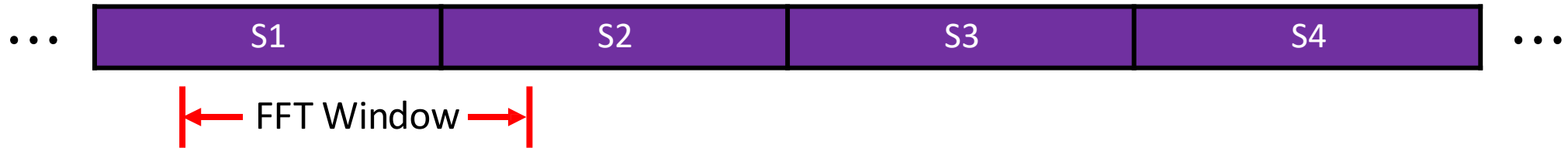
..., $+0.5 + 1i$, $-0.7 + 0.3i$, ...

✗ Cannot decode!

FFT window is misaligned with symbol

Subcarriers are no longer orthogonal.

OFDM Cyclic Prefix



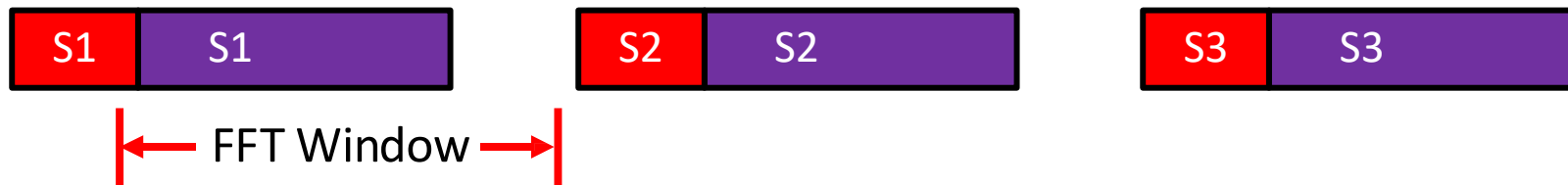
- DFT (FFT) assumes time samples are periodic of period N
- Circular Shift before taking FFT:

$$x[t] \rightarrow X[f]$$

$$x[t - \tau \bmod N] \rightarrow X[f] e^{-j \frac{2\pi f \tau}{N}}$$

Handwritten blue annotations: an arrow points from the τ in the time domain to the τ in the exponent of the frequency domain, and a wavy line is drawn under the exponential term.

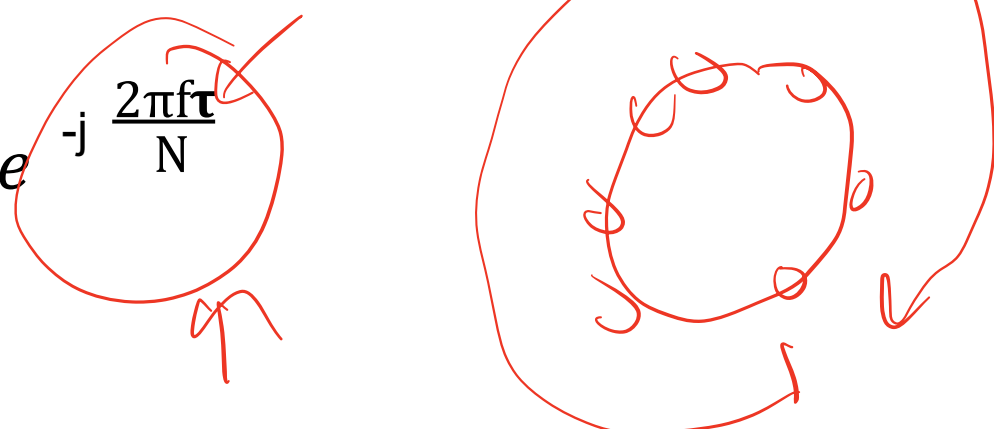
OFDM Cyclic Prefix



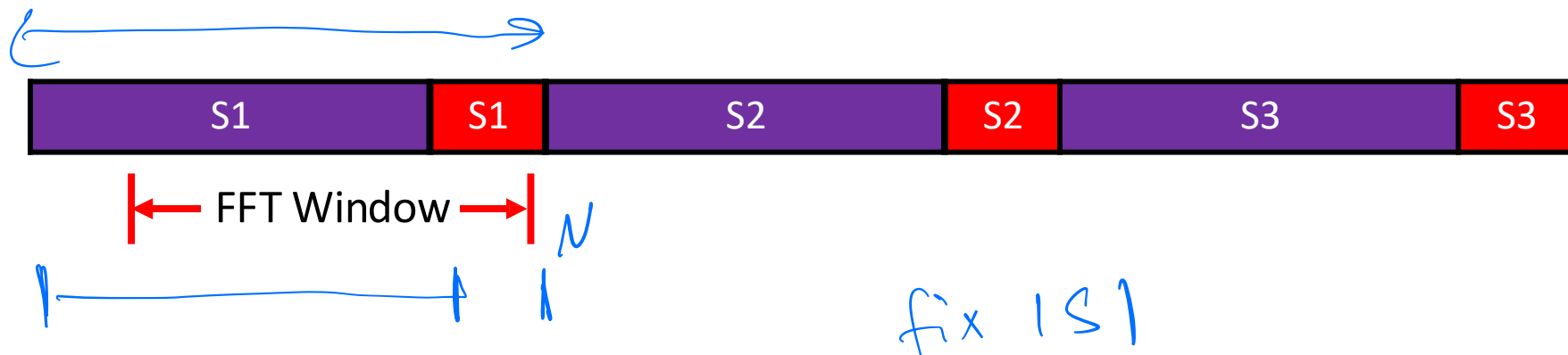
- DFT (FFT) assumes time samples are periodic of period N
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OFDM Cyclic Prefix

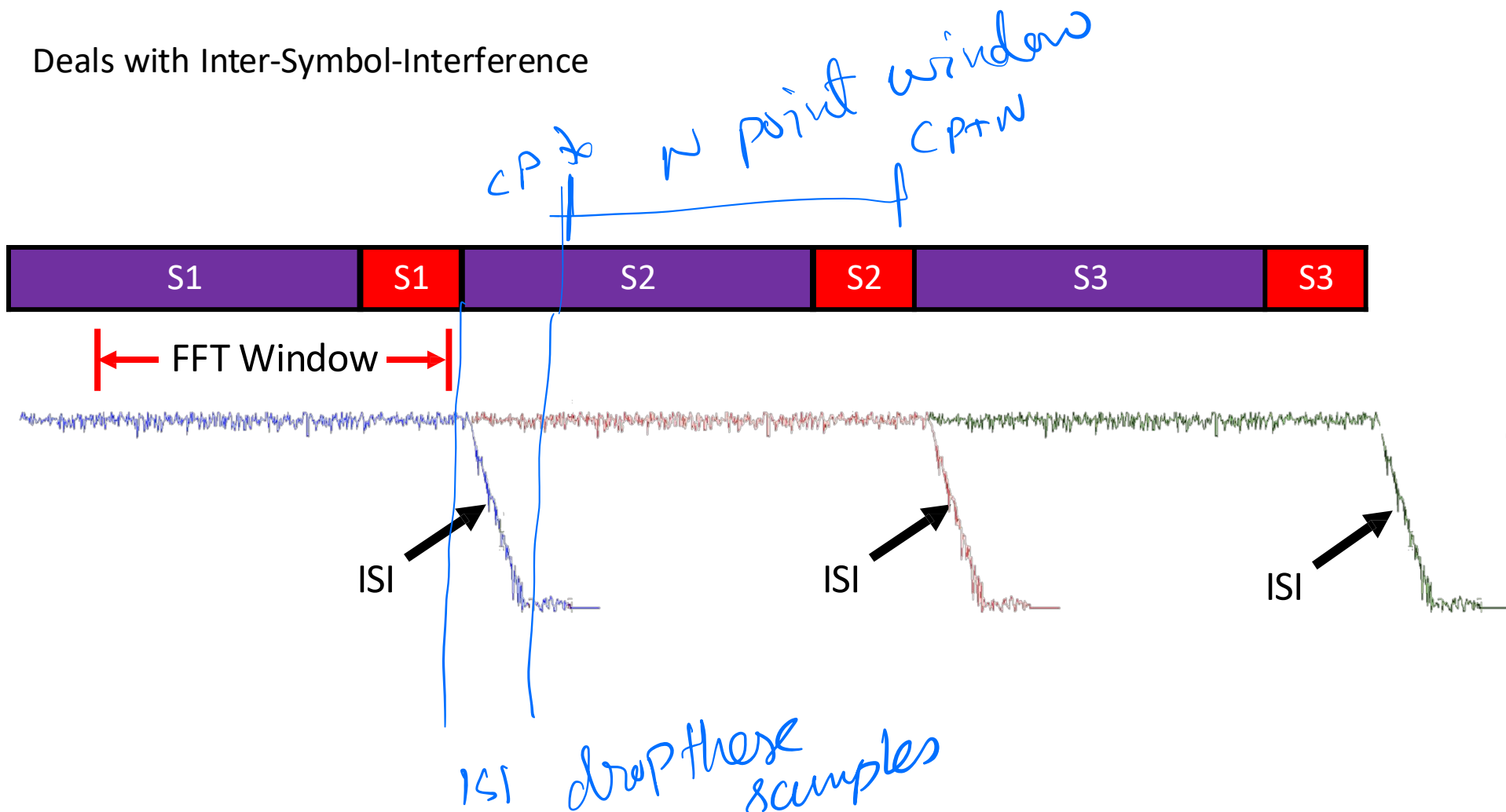


- Even if FFT window is misaligned, CP ensures that all samples come from the same symbol → Orthogonality is preserved!
- Simple Phase Shift → Can be corrected by lumping with channel $H[f]$
- Cyclic Prefix can be created by:
 - Take first few samples and append them to end of symbol.
 - Take last few samples and prefix them to beginning of symbol.

OFDM Cyclic Prefix

Cyclic Prefix:

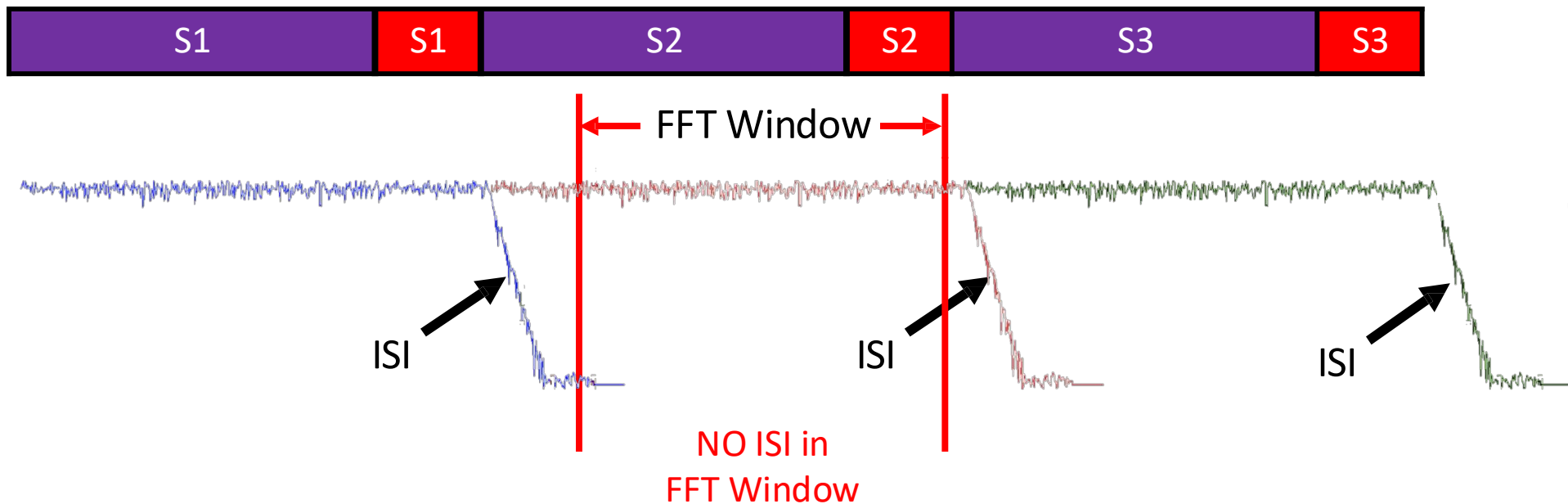
- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference



OFDM Cyclic Prefix

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference

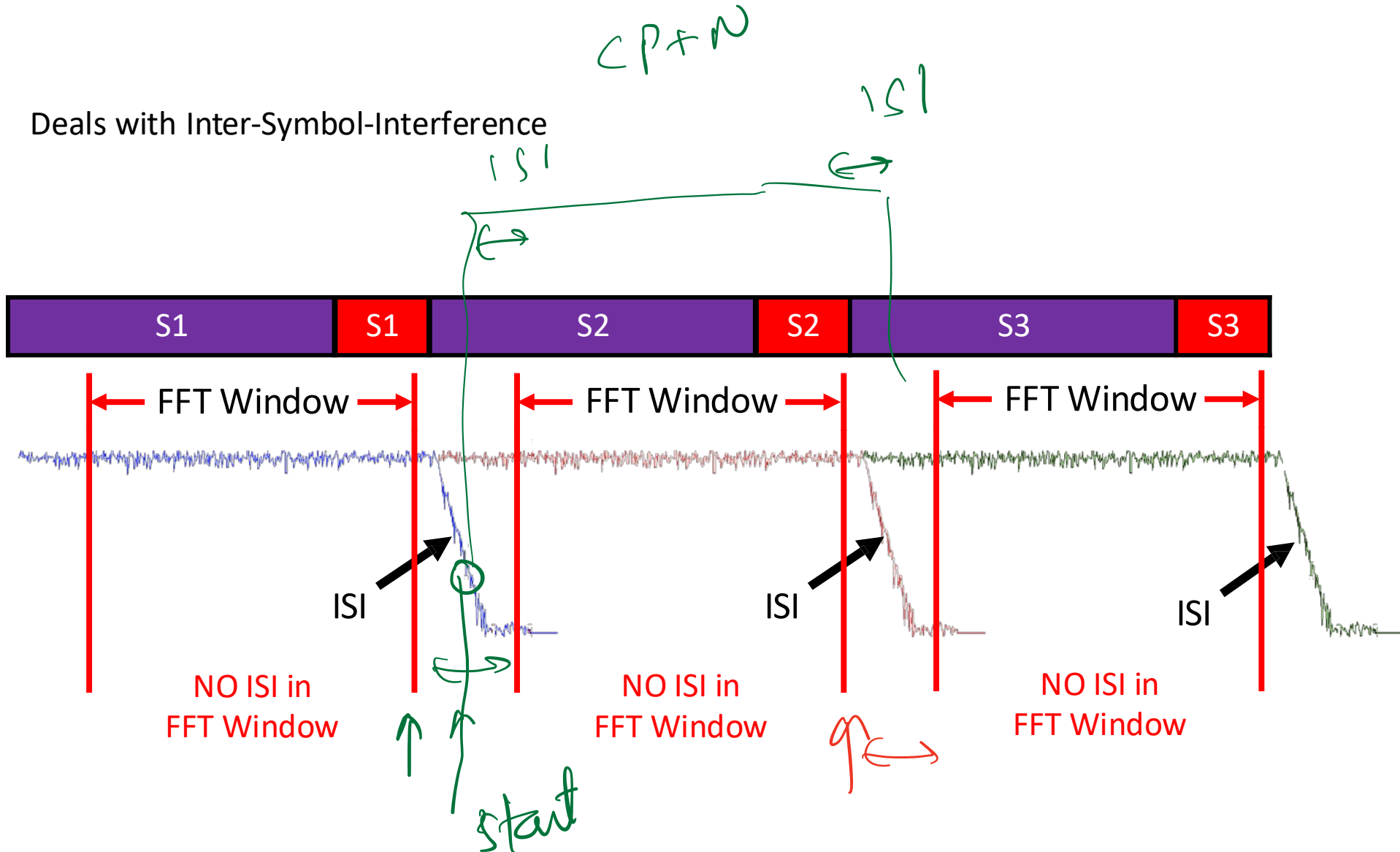


OFDM Cyclic Prefix

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window

- Deals with Inter-Symbol-Interference



OFDM Cyclic Prefix

Cyclic Prefix:

+ Preserves orthogonality by allowing some misalignment in FFT Window

+ Deals with Inter-Symbol-Interference

— Overhead: Send $CP + N$ samples for every N samples

$$\text{Overhead} = \frac{CP}{CP + N}$$

Handwritten notes: "extra" with an arrow pointing to CP, and "total" with an arrow pointing to CP + N.

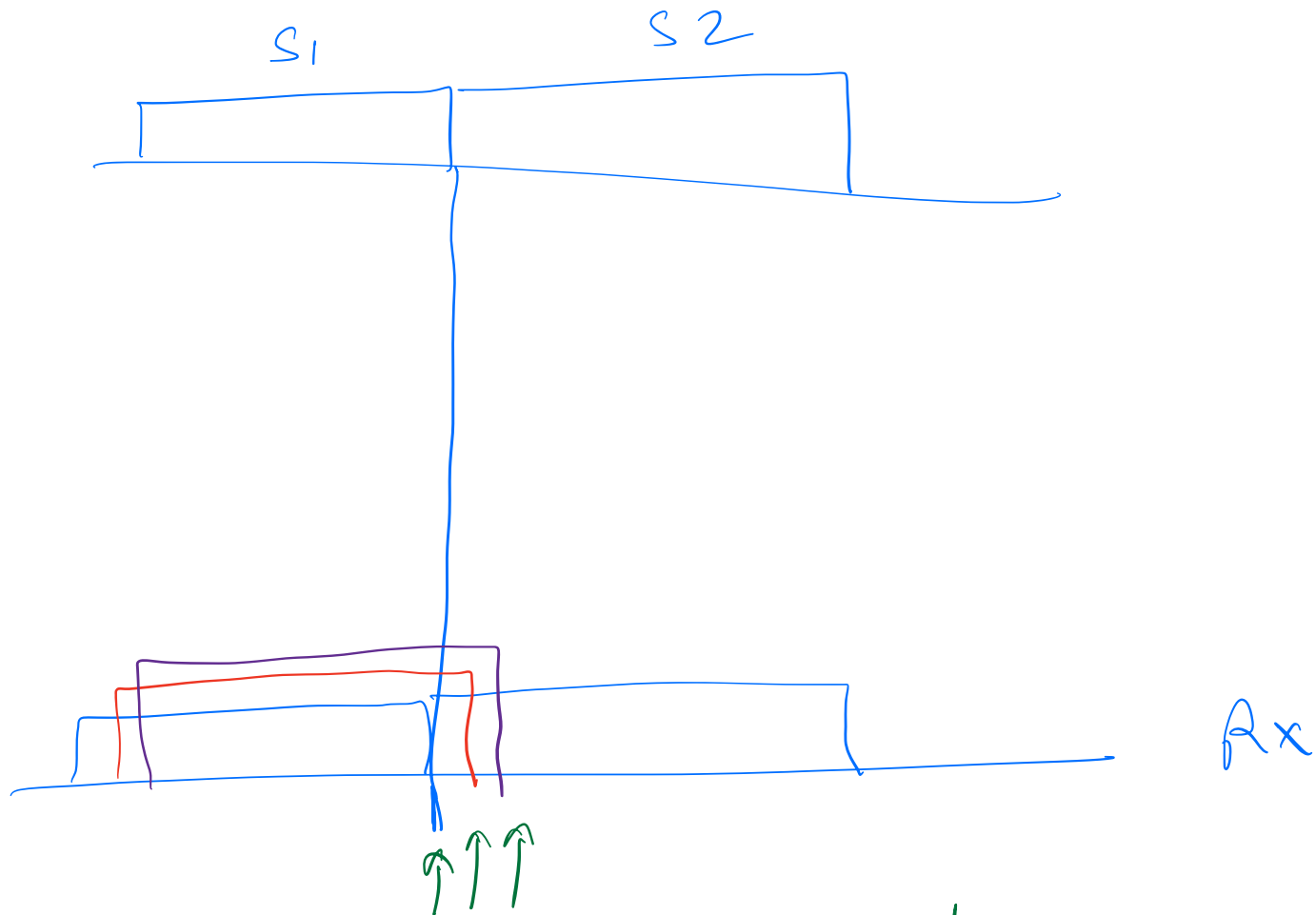
e.g. WiFi 802.11n: $N = 64$, $CP = 16 \rightarrow \text{Overhead} = 20\%$

e.g. LTE: $N = 1024$, $CP = 72 \rightarrow \text{Overhead} = 6.5\%$

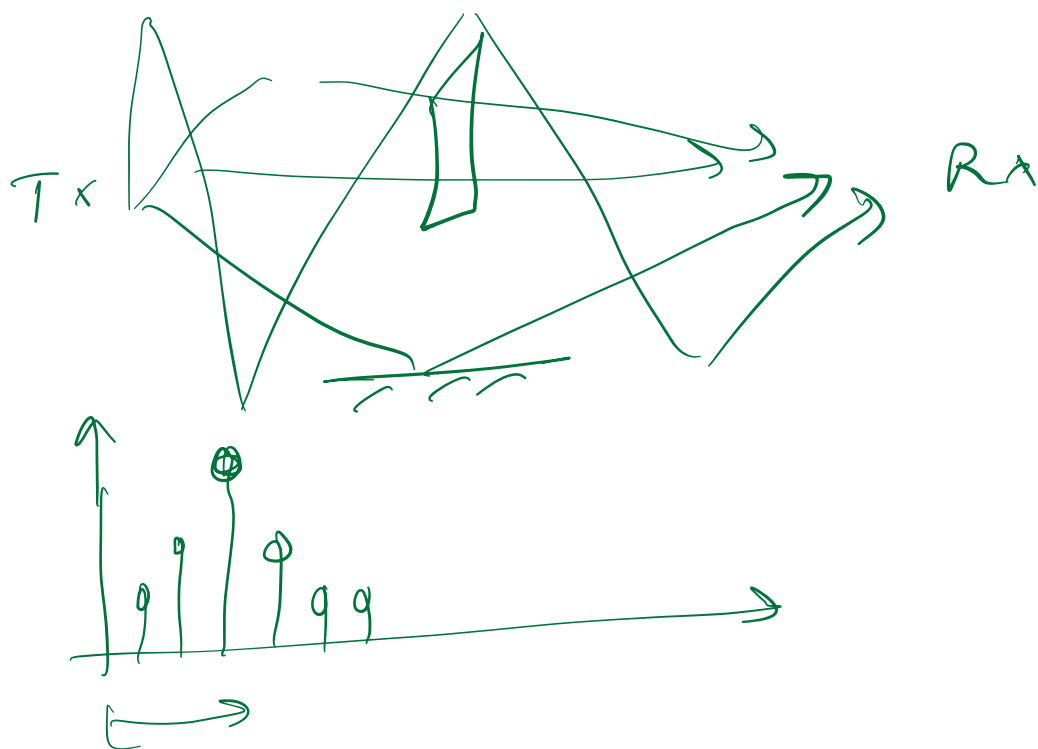
OFDM Cyclic Prefix



- Cyclic prefix is a not a bullet proof solution.
- Can still end up misaligned!
- Need a way to ensure we detect the beginning of the packet correctly.
- If we do, CP will ensure that even if we are not accurate, we can still decode.



$\uparrow$ first copy $\Rightarrow$ good
 $\Rightarrow$ generally strongest
 $\Rightarrow$ may not be true



OFDM Packet Detection

- Detect Beginning of packet to make sure we are within the CP

$(N + CP)$

- Send Training Sequence: Preamble Symbols

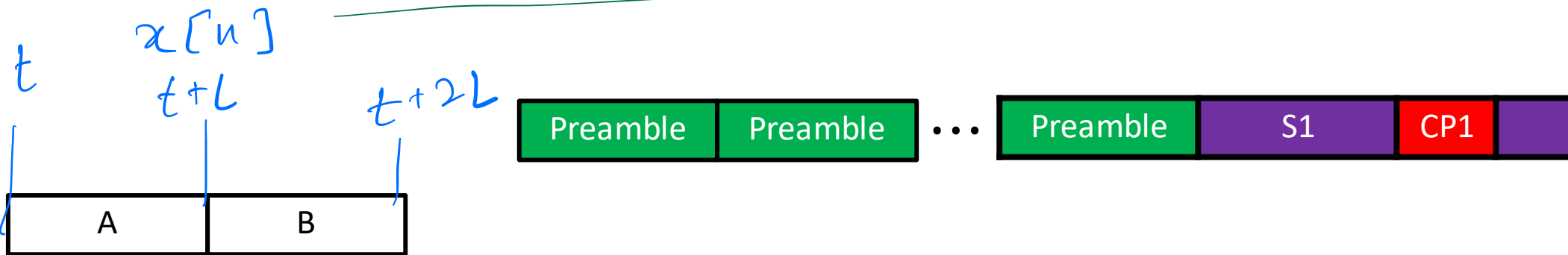
- Preamble Symbols: Known Symbol Repeated at the beginning of packet



LTS: short time seq. → repeated 10 times in wifi.

- Generally, 2 copies of preamble (LTS – Long Training Sequence) used in WiFi + 0.5 copy as cyclic prefix

OFDM Packet Detection: Sliding Window



- Two windows of L ($2N$) samples each.

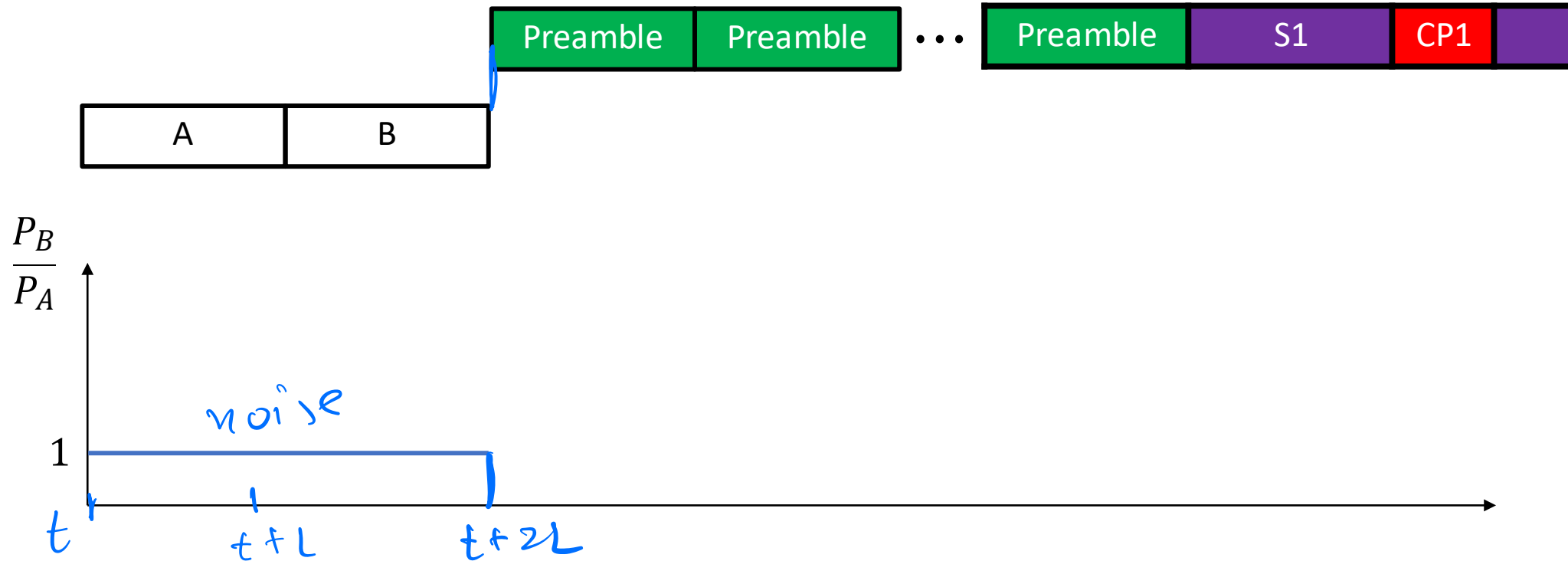
- Compute:

$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

energy in B

energy in A

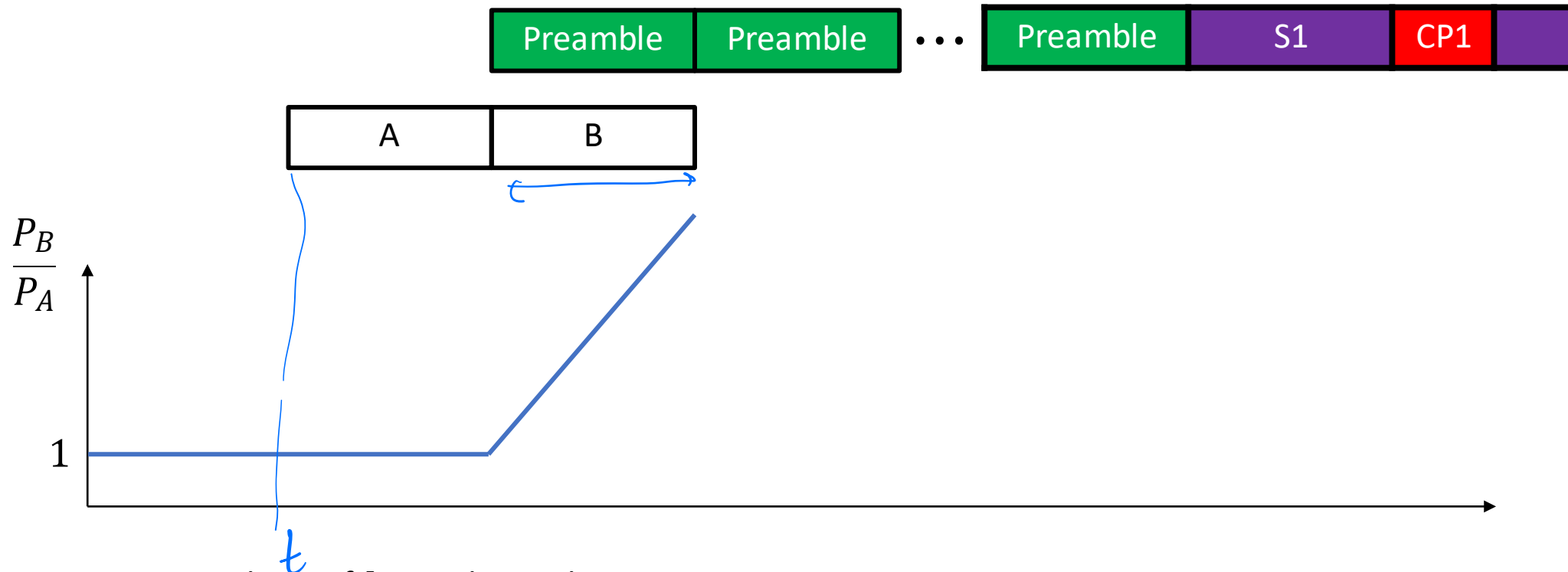
OFDM Packet Detection: Sliding Window



- Two windows of L ($2N$) samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

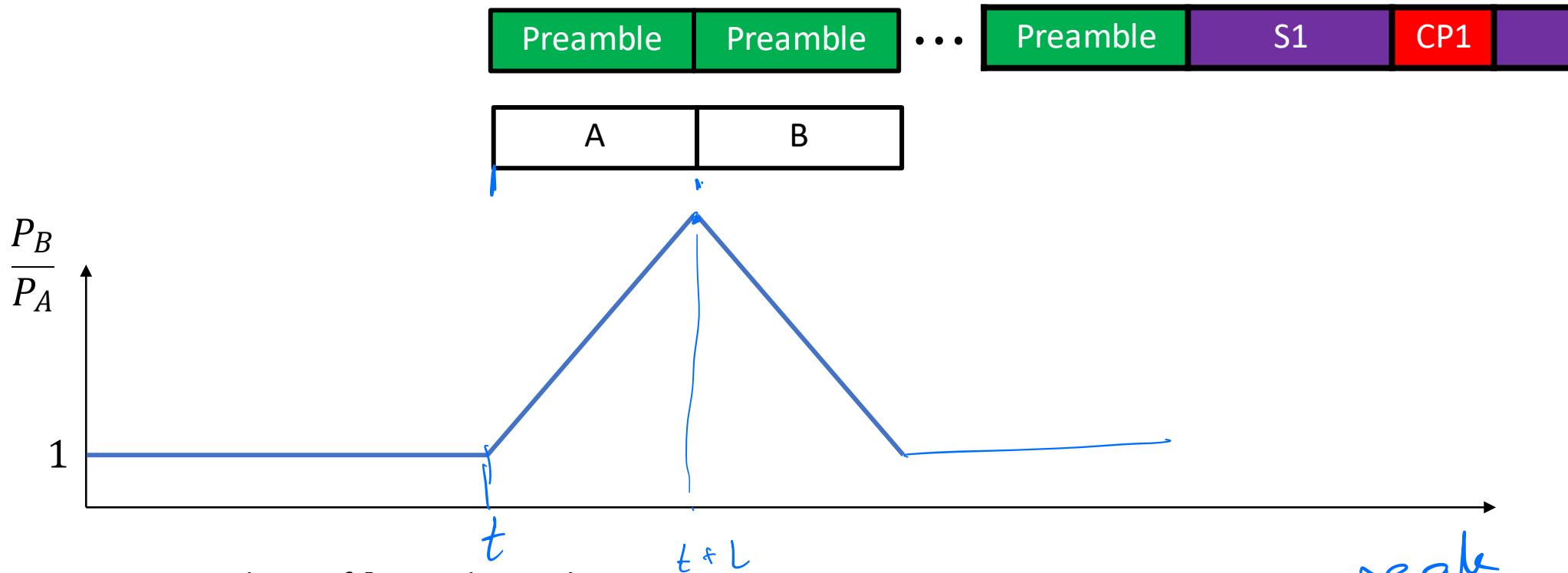
OFDM Packet Detection: Sliding Window



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- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

OFDM Packet Detection: Sliding Window

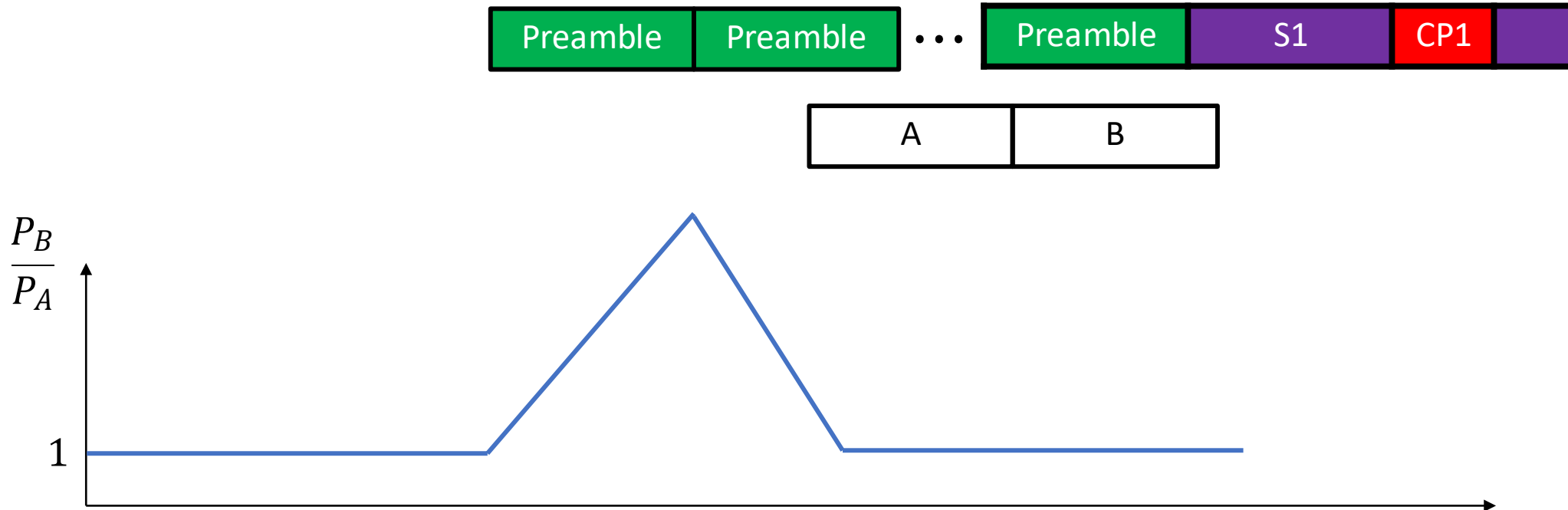


- Two windows of L samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

*find t where peak appears
 $\Rightarrow$ find appropriate start of packet.*

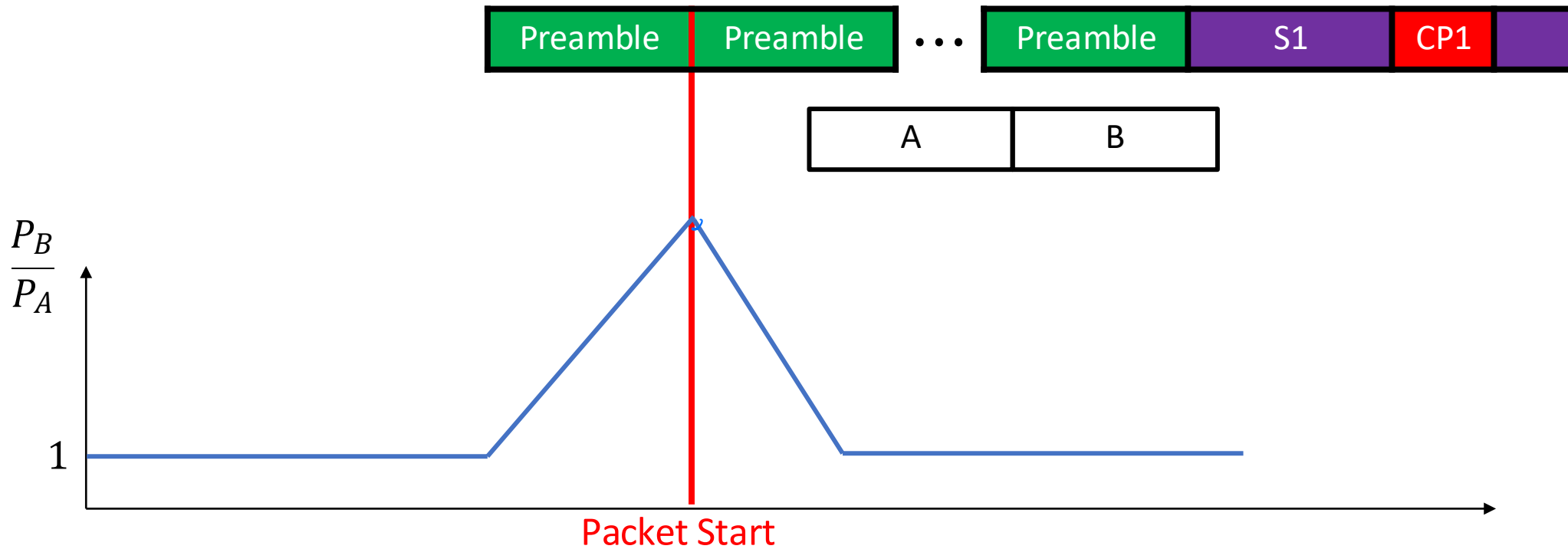
OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

OFDM Packet Detection: Cross Correlation

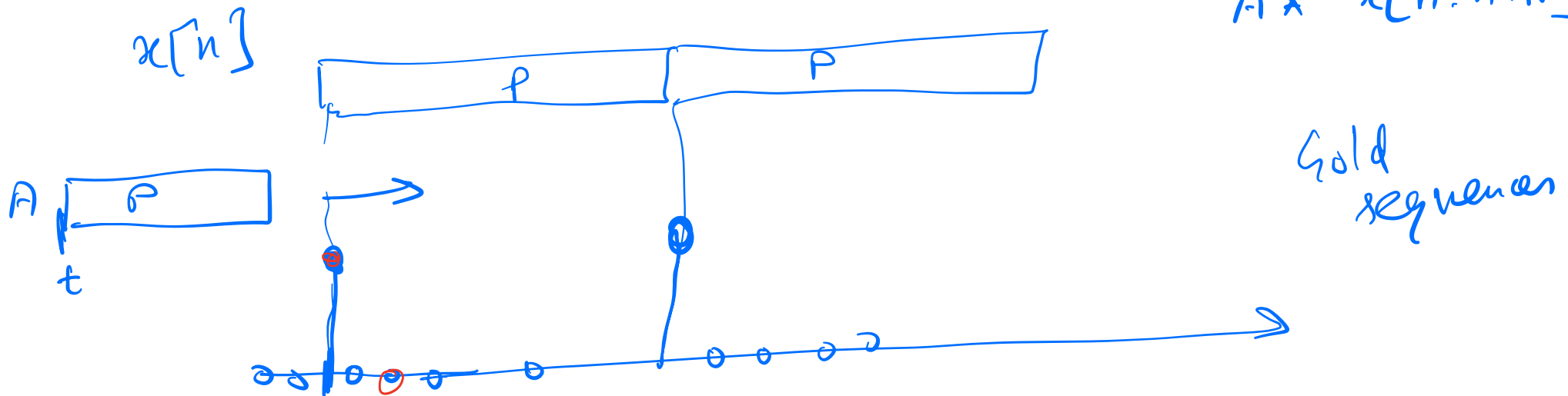
Robust - preamble uses PN sequence
(pseudonoise)
with interesting properties

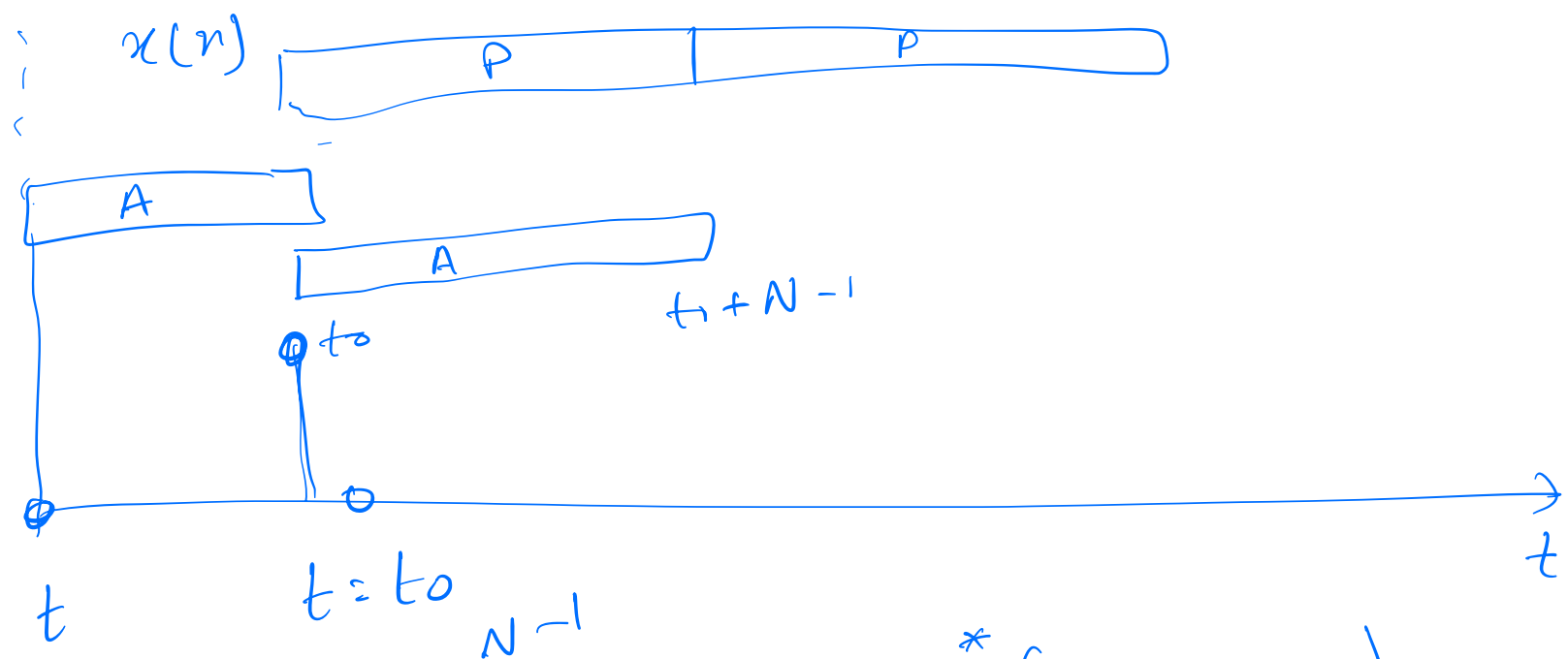
Preamble $P(n)$

$$|P(n) * P(n)| = 1$$

$$|P(n-k) * P(n)| = 0$$

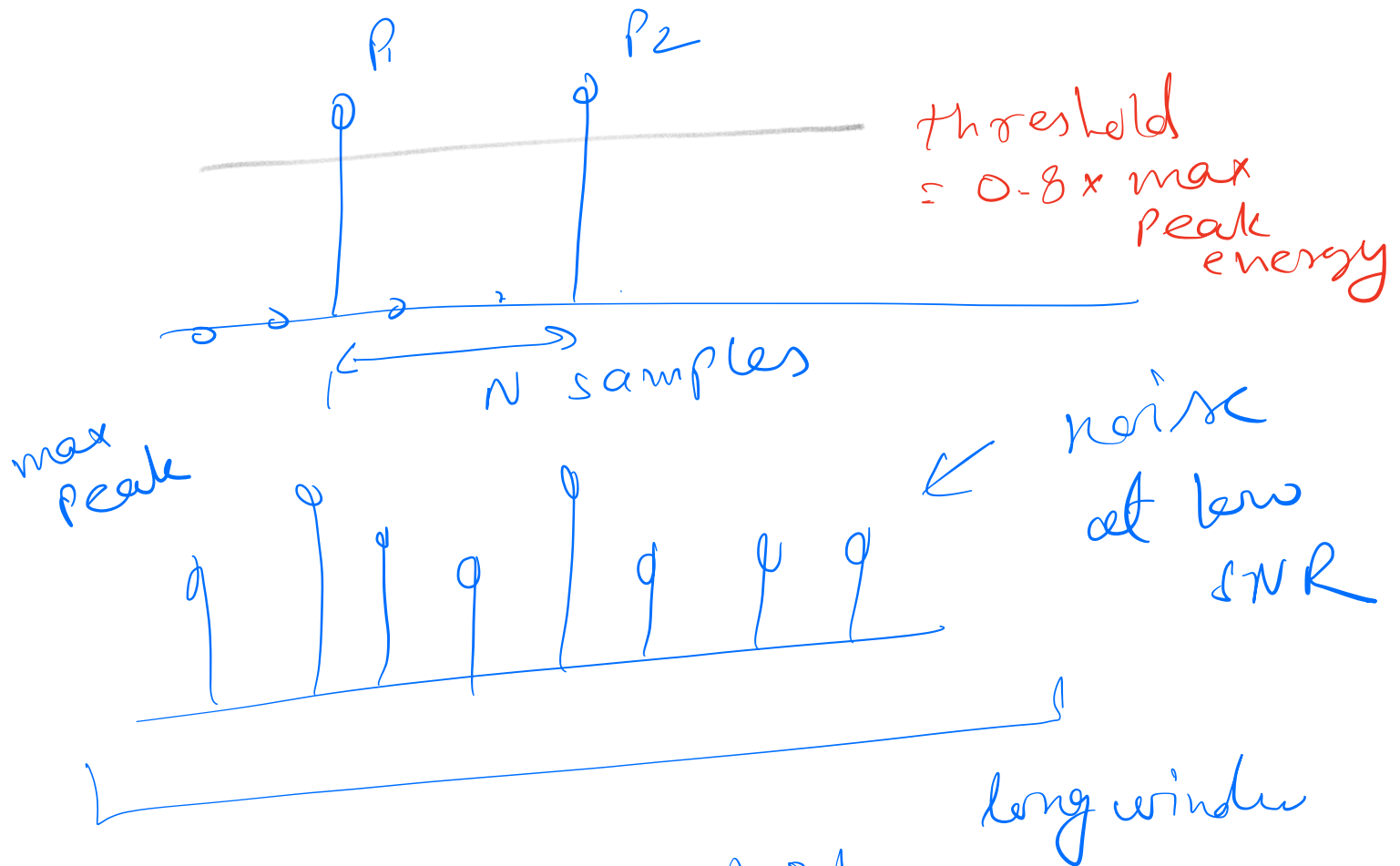
$$A * x[n:n+N]$$





$$R(t) = \sum_{k=0}^{N-1} A(k) x^*(t+k)$$

Here, A is a copy of same preamble known to the Rx



find two peaks which are exactly shifted by N samples

$\Rightarrow$ more robust to noise
 $\Rightarrow$ more robust to CFO freq. offset

Multipath

Preamble 1

Preamble 2

$R(t)$

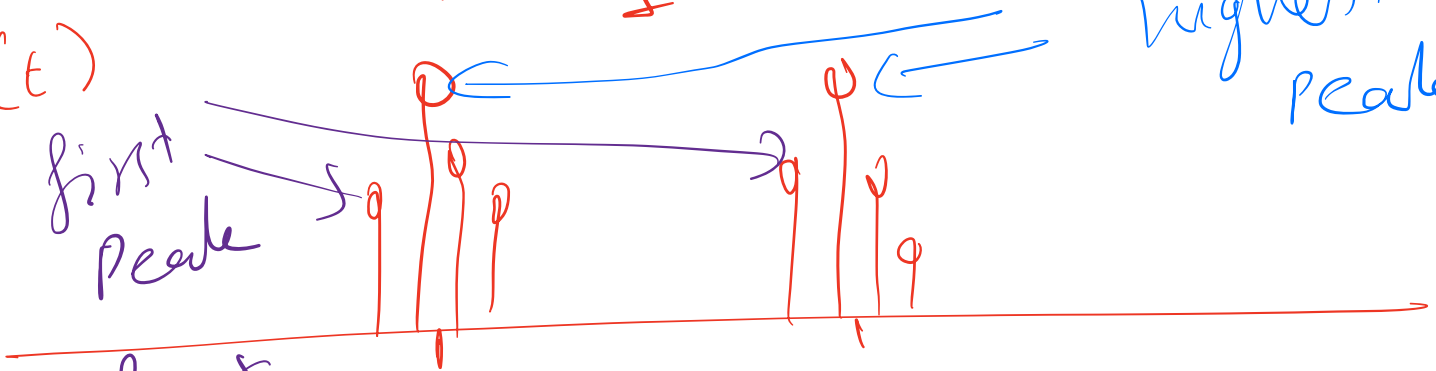
first
Peak

highest
Peak

$\Rightarrow$ first
path

$\Rightarrow$ least
traveled
path.

N samples



Optimal OFDM Packet Detection: Schmidl-Cox, 802.11 style (Reading Exercise - Piazza)

receive

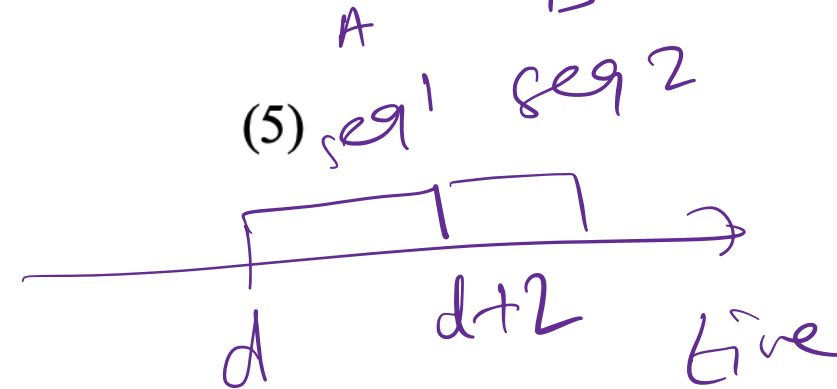
=> auto correlation.

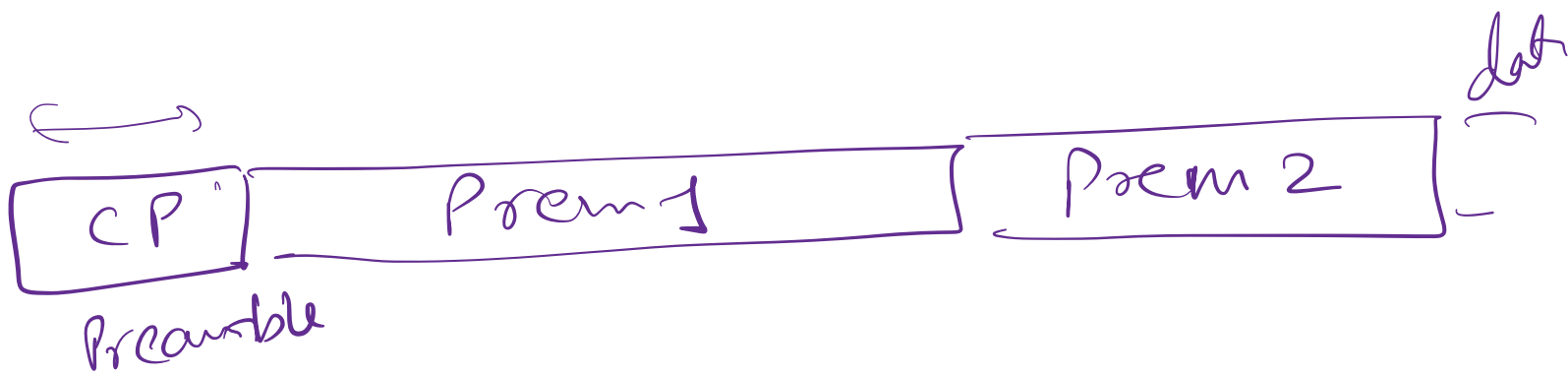
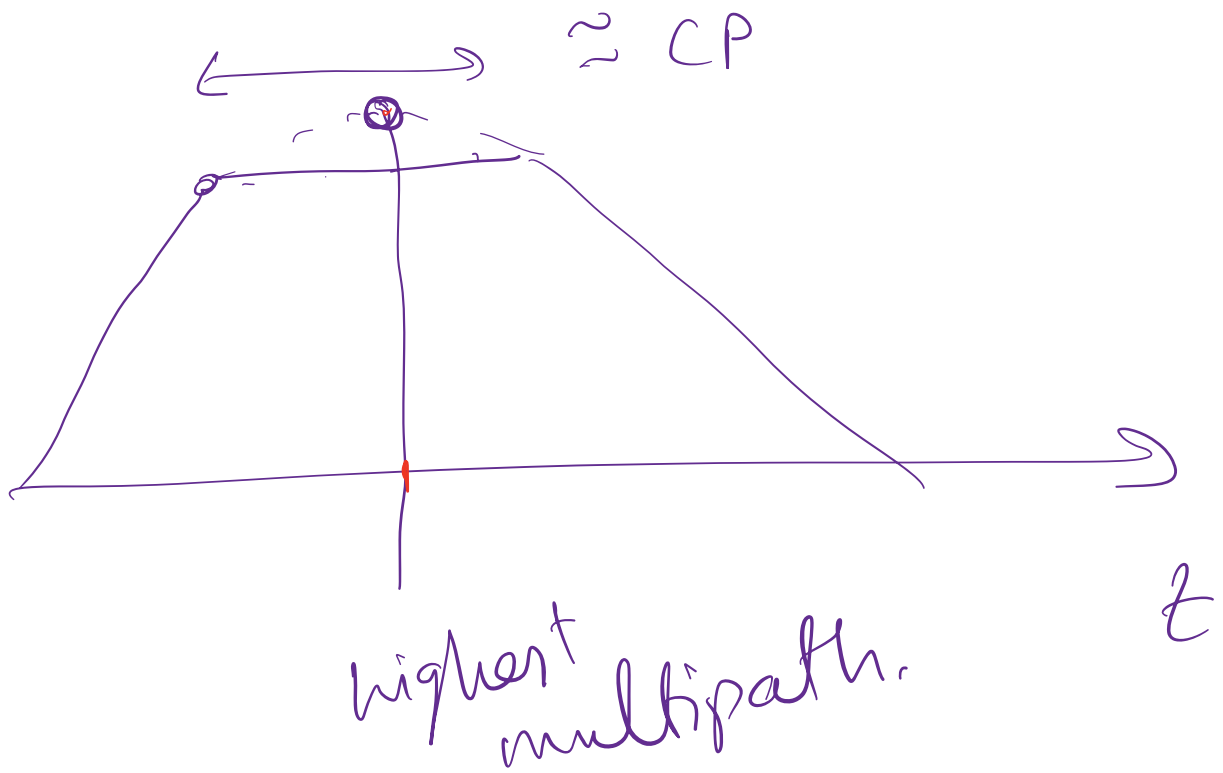
$$P(d) = \sum_{m=0}^{L-1} (r_{d+m}^* r_{d+m+L})$$

$$R(d) = \sum_{m=0}^{L-1} |r_{d+m+L}|^2$$

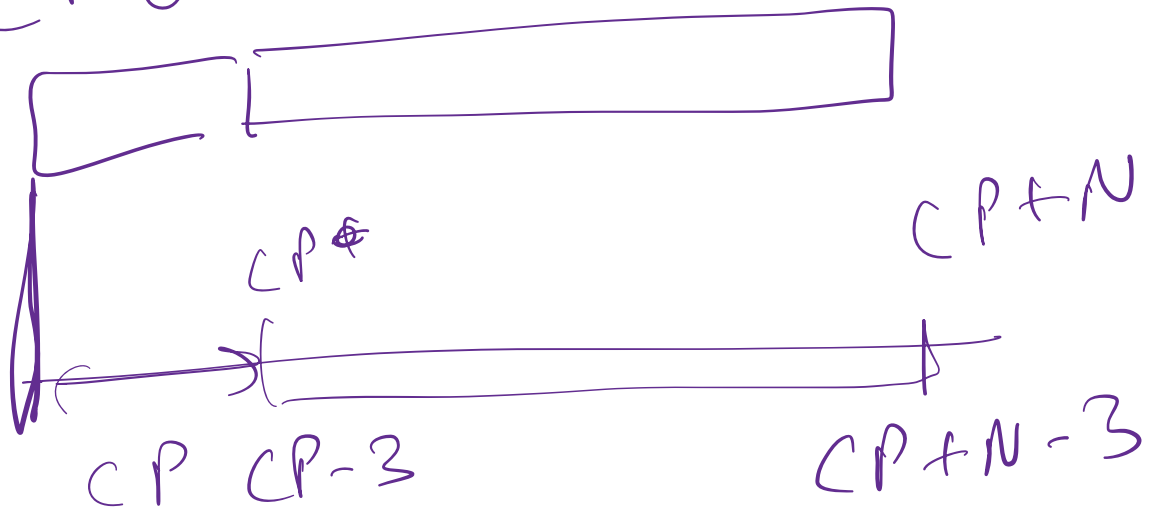
$$M(d) = \frac{|P(d)|^2}{(R(d))^2}$$

den = normalization.





CP data data



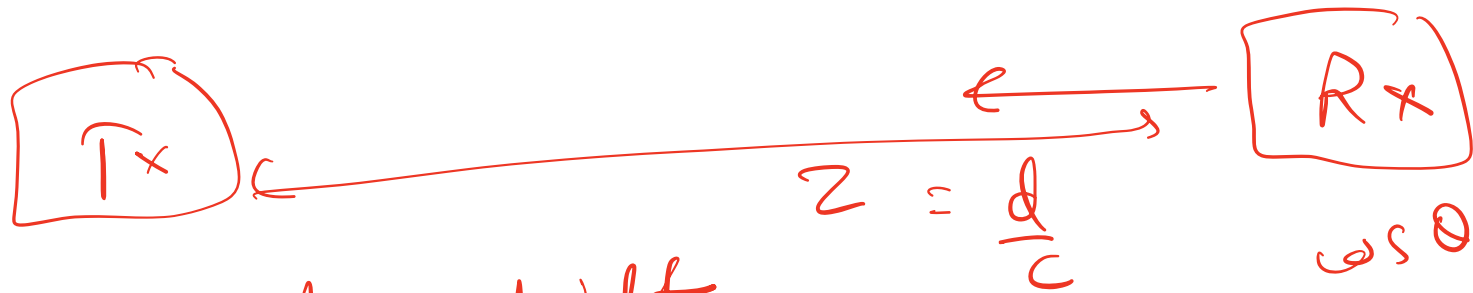
time

How often we send preamble over time?

Ans: Channel coherence time

mobility

velocity $v \cos \theta$



Doppler shift

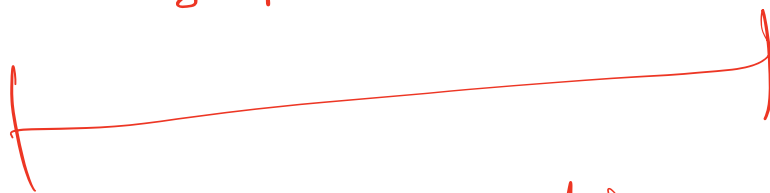
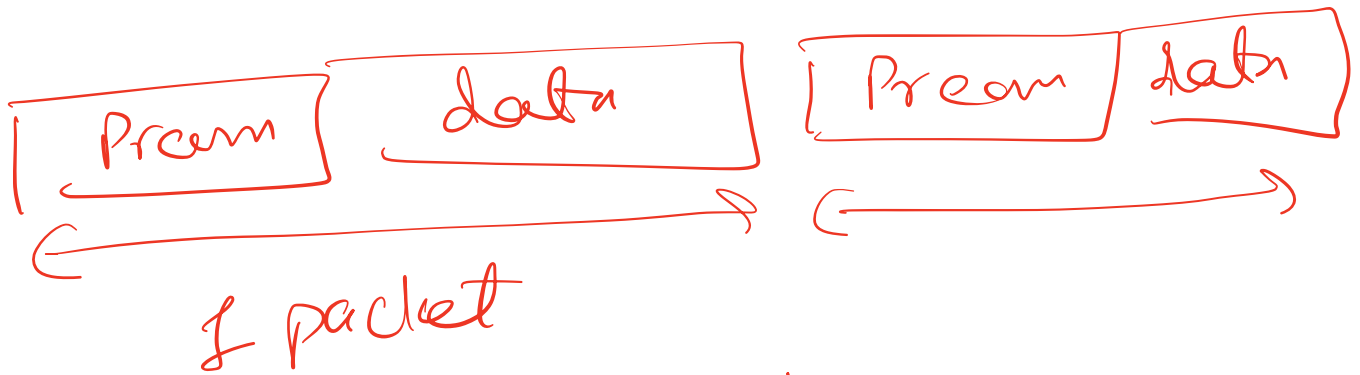
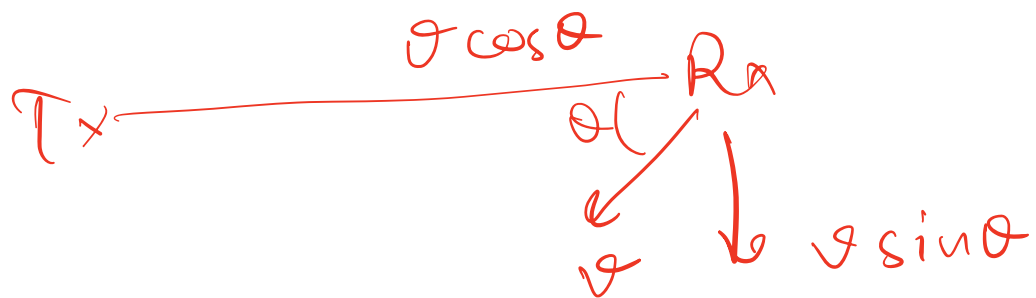
$$= \frac{d_0 + vt}{c}$$

$$2\pi f_c z =$$

$$2\pi f_c \left(\frac{d_0 + vt}{c} \right) =$$

$$2\pi f_c \frac{d_0}{c} + 2\pi f_c \frac{vt}{c}$$

$$f_d = \frac{f_c v \cos \theta}{c} \quad \text{Doppler freq}$$



time = decided by
channel
coherence
time

WiFi Example: STS and LTS

wifi 802.11n

[guard interval]

2.5 copies

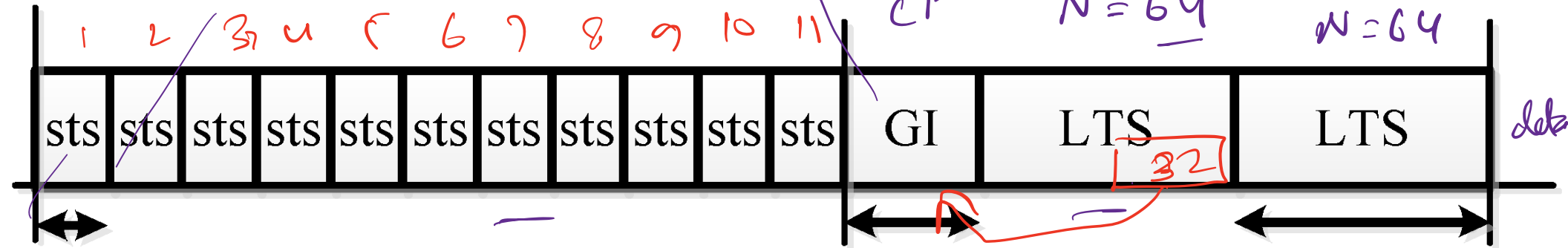
32 length

CP

$N = 64$

Preamble

$N = 64$



0.8 μs

1.6 μs

auto correlation here

3.2 μs

↑ start of data

11 copies

⇒ sliding window packet detection

⇒ ACL = Automatic gain control

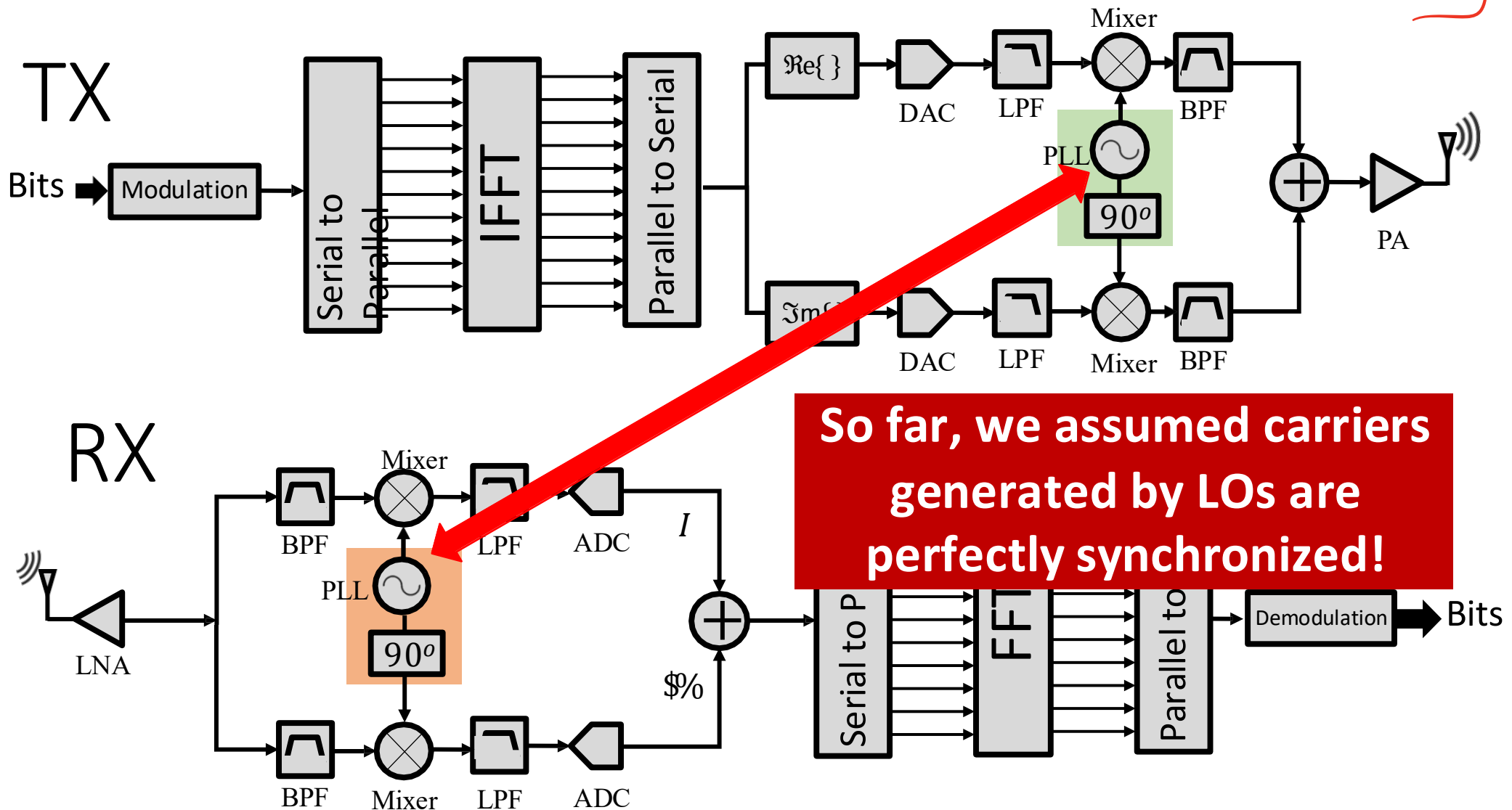
$f_s = 20 \text{ MHz}$

$T_s = \frac{1}{f_s} = 0.05 \text{ μs}$

$N T_s = 64 \times 0.05 = 3.2 \text{ μs}$

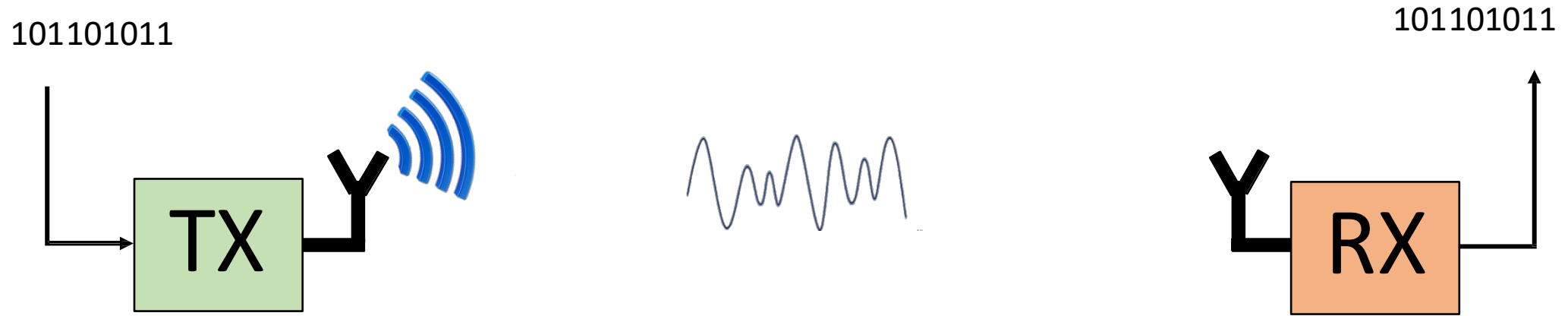
OFDM: Orthogonal Frequency Division

Multiplexing Transmit Symbols in Frequency Domain On Orthogonal Subcarriers



So far, we assumed carriers generated by LOs are perfectly synchronized!

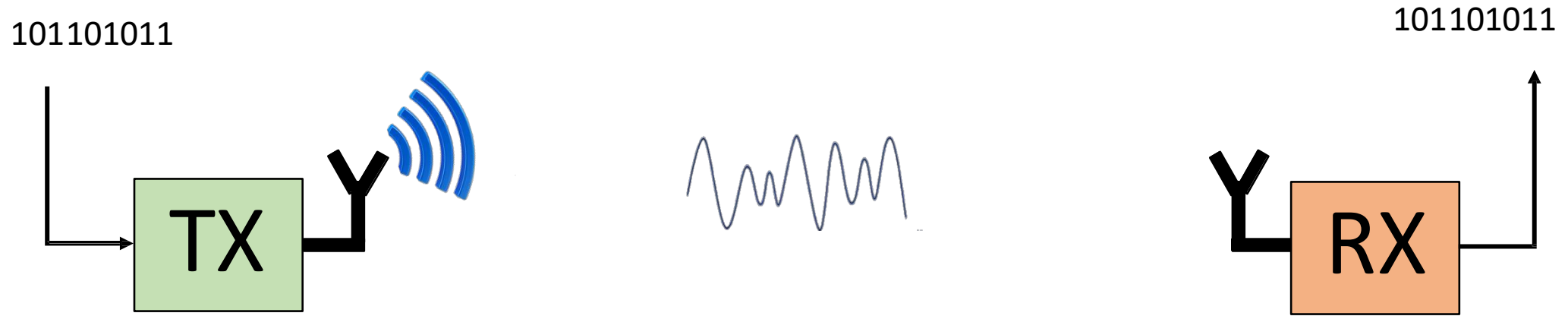
Carrier Frequency Offset



$$\begin{aligned}
 x(t) &\Rightarrow x(t) \times e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \times e^{j2\pi f_c t} \\
 &\Rightarrow h(t) * x(t) \\
 &\Rightarrow y(t) = h(t) * x(t) + v(t)
 \end{aligned}$$

Assumes TX & RX perfectly synched

Carrier Frequency Offset



$$x(t) \Rightarrow x(t) \times e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \Rightarrow h(t) * x(t) e^{-j2\pi f_c t} \times e^{j2\pi f'_c t}$$

TX & RX are not synched

$$\Rightarrow h(t) * x(t) e^{-j2\pi \Delta f_c t}$$

$$\text{CFO: } \Delta f_c = f_c - f'_c$$

$$\Rightarrow y(t) = h(t) * x(t) e^{-j2\pi \Delta f_c t} + v(t)$$

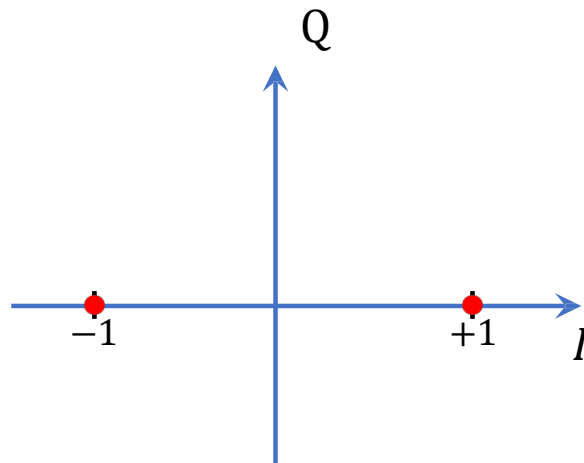
Phase changes with time!

Carrier Frequency Offset

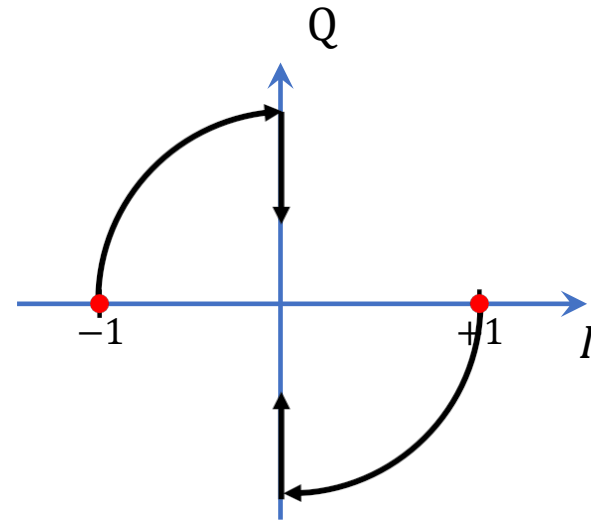
Consider BPSK Modulation.

$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$



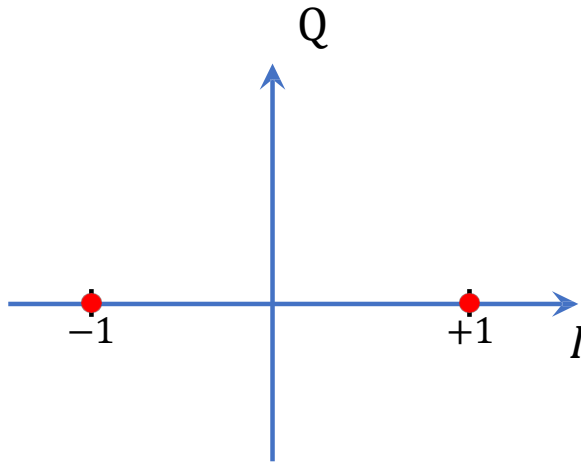
$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$

Carrier Frequency Offset

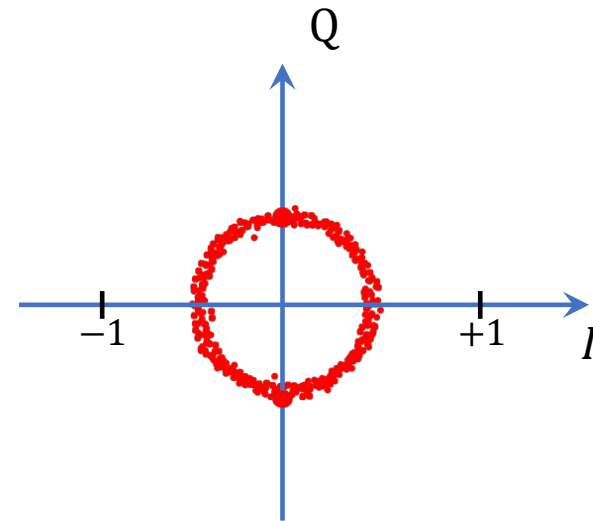
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$x(t)$

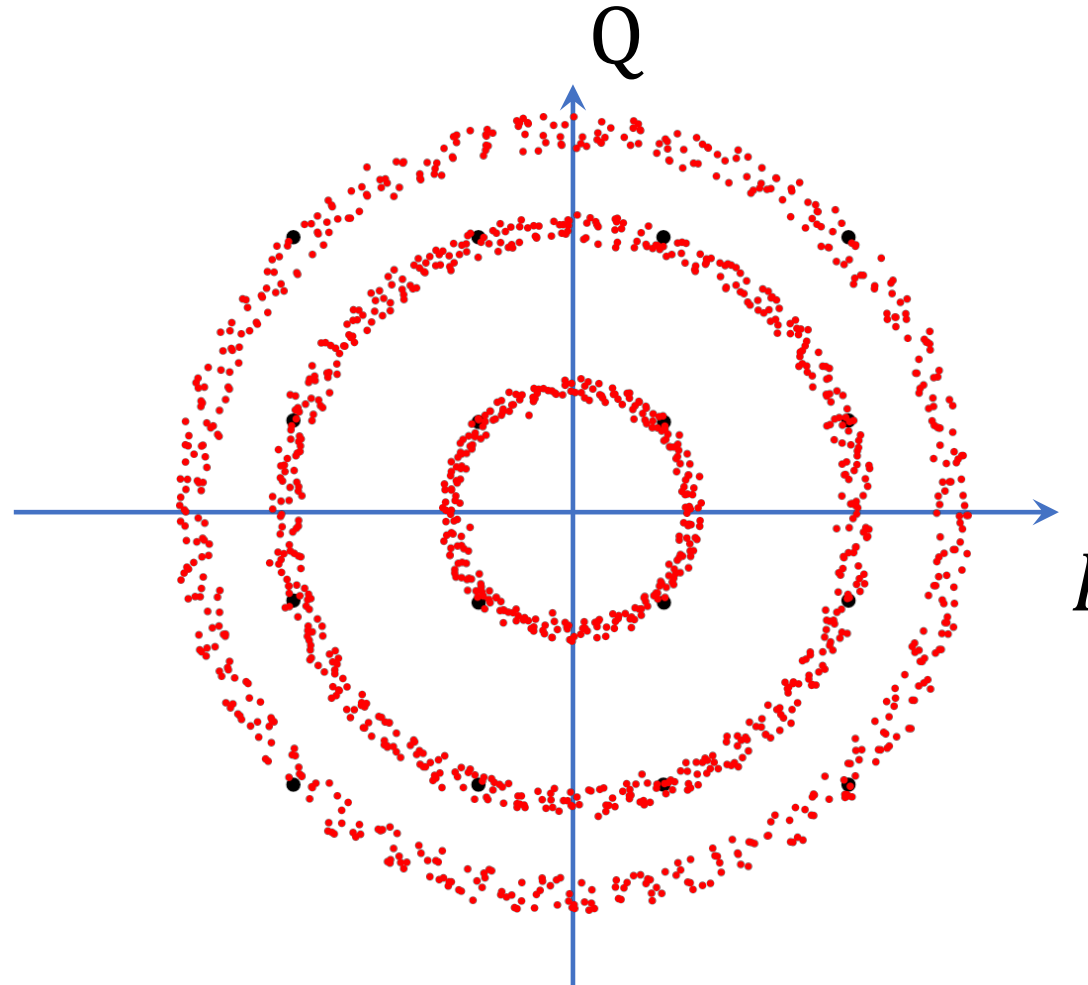


$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$

Impossible to Decode!

Carrier Frequency Offset

Consider 16 QAM Modulation



Need to estimate and correct CFO to decode!

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1(t) = x(t)e^{-j2\pi\Delta f_c t}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

$$y_2(t) = x(t)e^{-j2\pi\Delta f_c (t + T)}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

- Compute:
$$A = \sum_{t=1}^N y_1[n]y_2^*[n] = \sum_{t=1}^N x[n]x^*[n]e^{j2\pi\Delta f_c NT_s}$$

$$= e^{j2\pi\Delta f_c NT_s} \sum_{t=1}^N |x[n]|^2 \quad \rightarrow \quad \Delta f_c = \frac{\angle A}{2\pi NT_s}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

- Compute:
$$A = \sum_{t=1}^N y_1[n]y_2^*[n] \quad \longrightarrow \quad \Delta f_c = \frac{\angle A}{2\pi nT_s}$$

- Correct CFO:
$$y[n] \times e^{j2\pi\Delta f_c nT_s}$$

OFDM Channel Estimation



- Use Preamble to estimate the channel

$$y(t) = h(t) * x(t) \leftrightarrow Y(f) = H(f) X(f)$$

- Send $X(f)$: $-1, +1, -1, -1, -1, +1, \dots$
- Receive: $-H(1), H(2), -H(3), -H(4), -H(5), H(6), \dots$

- Estimate: $\tilde{H}(f) = \frac{Y(f)}{X(f)}$

- Use two preambles to average noise: $\tilde{H}(f) = \frac{Y_1(f) + Y_2(f)}{2 X(f)}$

Phase Tracking

So Far: Estimated and Corrected For Coarse Value of CFO

- Residual CFO:

$$y(t) = h(t) * x(t) e^{j2\pi\Delta f_c t} + v(t)$$

$$\Delta f_c = df_c + \delta f_c$$

Coarse CFO

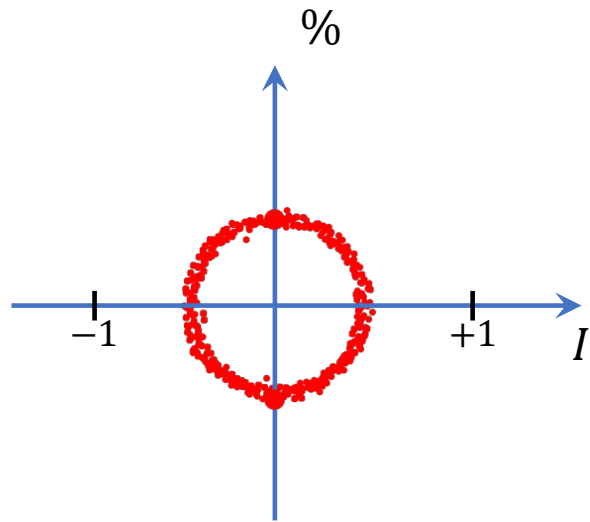
Residual CFO

We estimated and corrected for coarse CFO!

Even small residual can accumulate over time
to create large phase: $e^{-j2\pi\delta f_c t}$

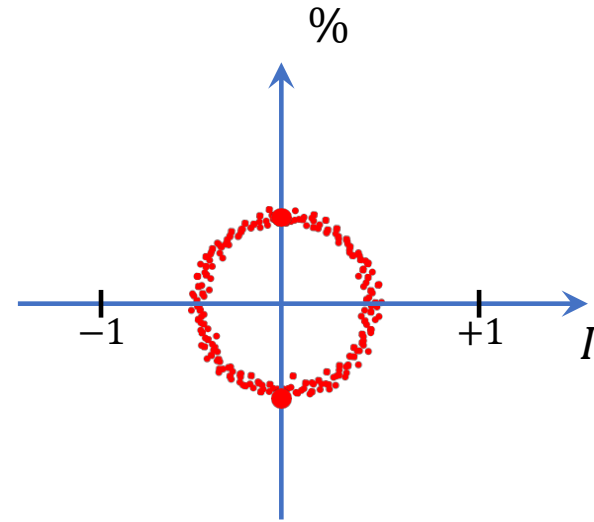
Need to track the phase

Phase Tracking



$$h x(t - r) e^{j2\pi\Delta f_c t} + v(t)$$

Without Course CFO Correction



$$h x(t - r) e^{j2\pi\delta f_c t} + v(t)$$

Without Residual CFO Correction

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

$$y[n] = x[n + n\delta T_s] e^{-j2\pi\delta f_c n T_s}$$

$$Y[f] = X[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi\delta f_c n T_s}$$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

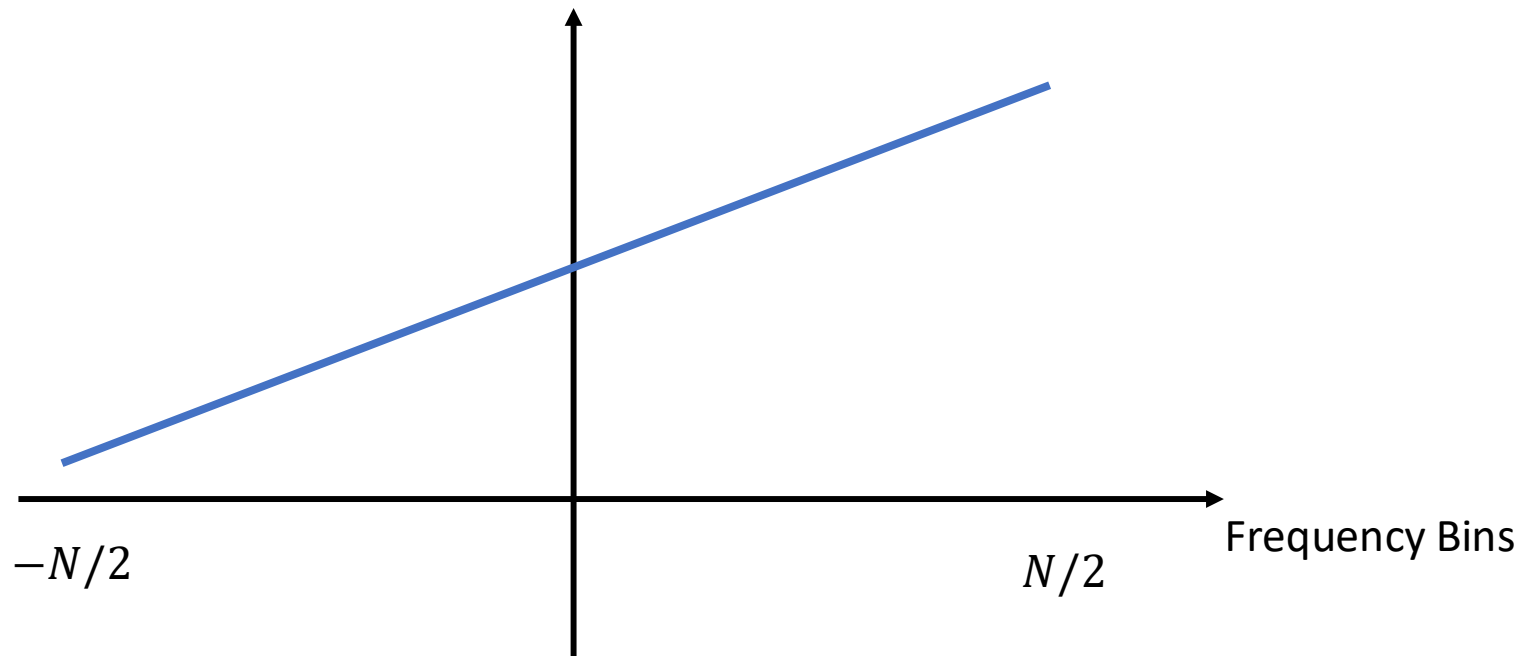
$$Y_1[f] = X_1[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi \delta f_c n T_s}$$

$$Y_2[f] = X_2[f] e^{j \frac{2\pi f (n+N+CP) \delta T_s}{N} - 2\pi \delta f_c (n+N+CP) T_s}$$

Phase accumulation: $\Delta\phi = 2\pi f \frac{(N+CP)\delta T_s}{N} - 2\pi \delta f_c (N+CP) T_s$

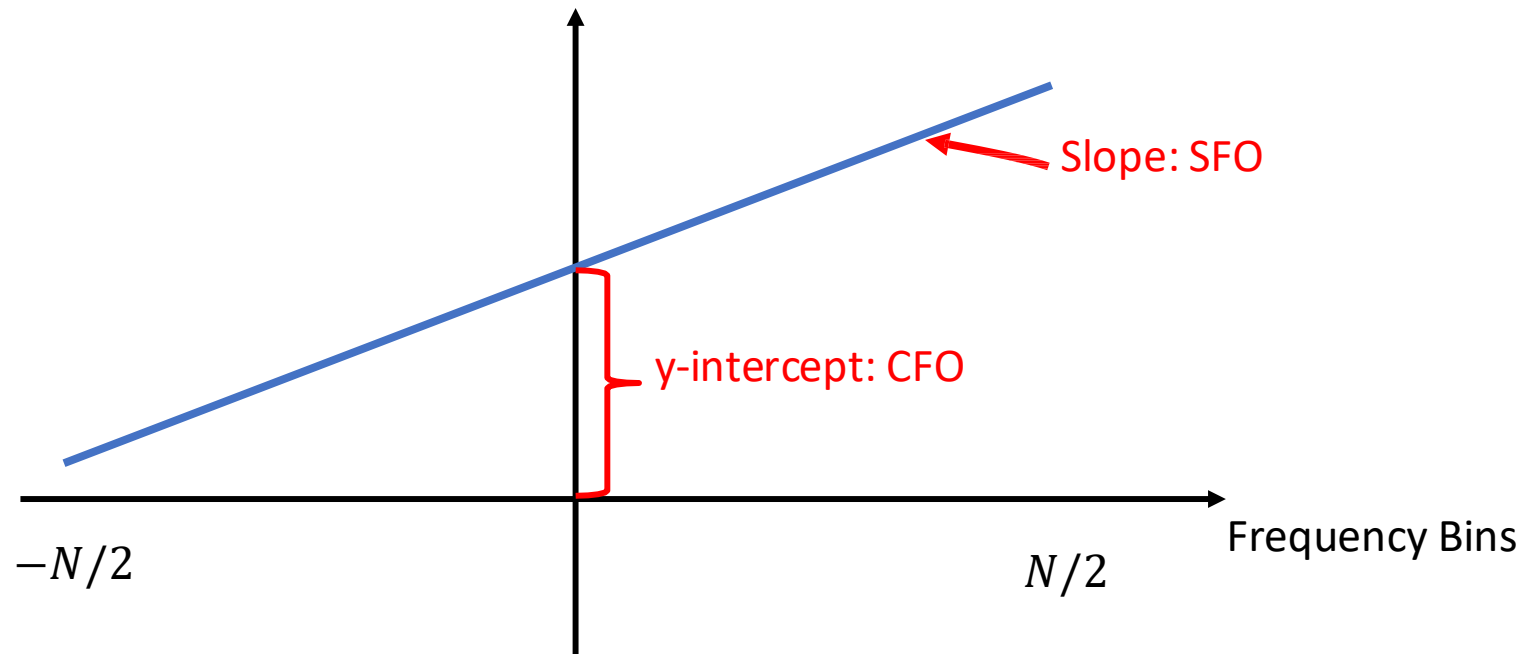
Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)
- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



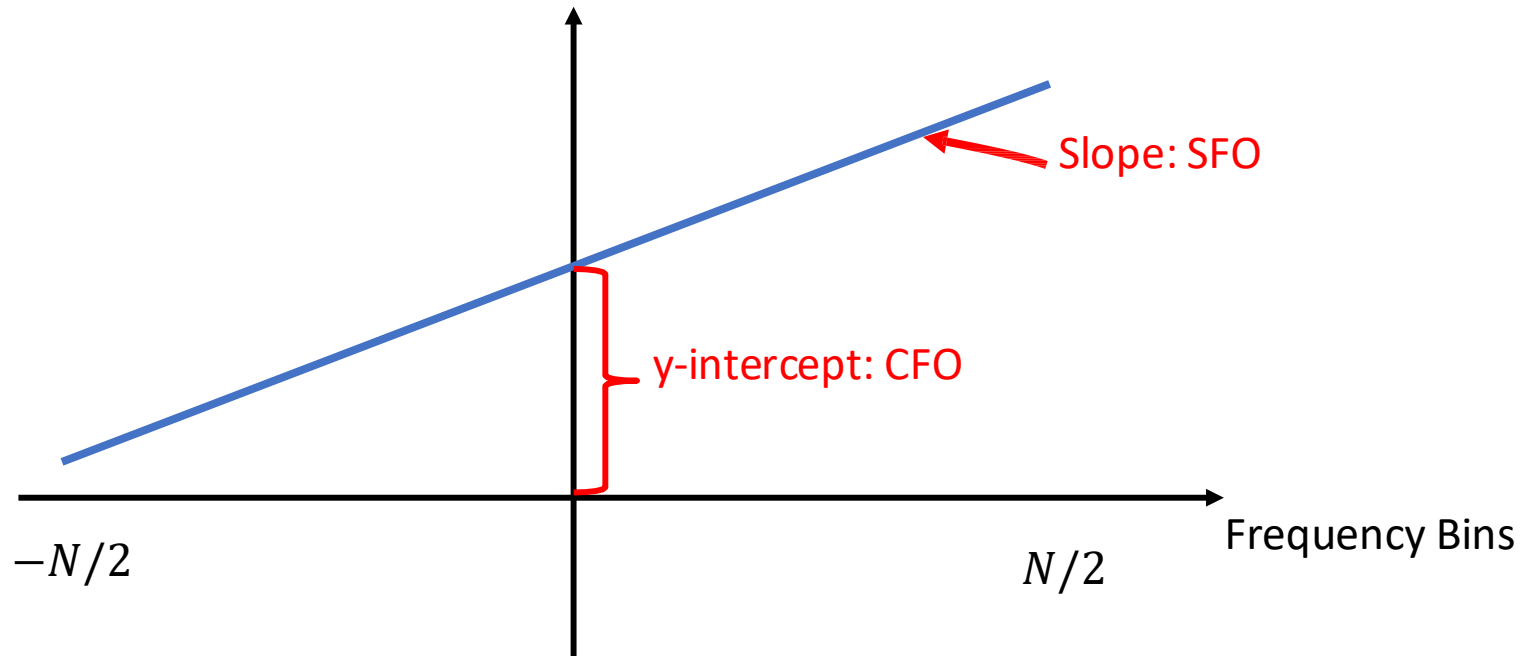
Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)
- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



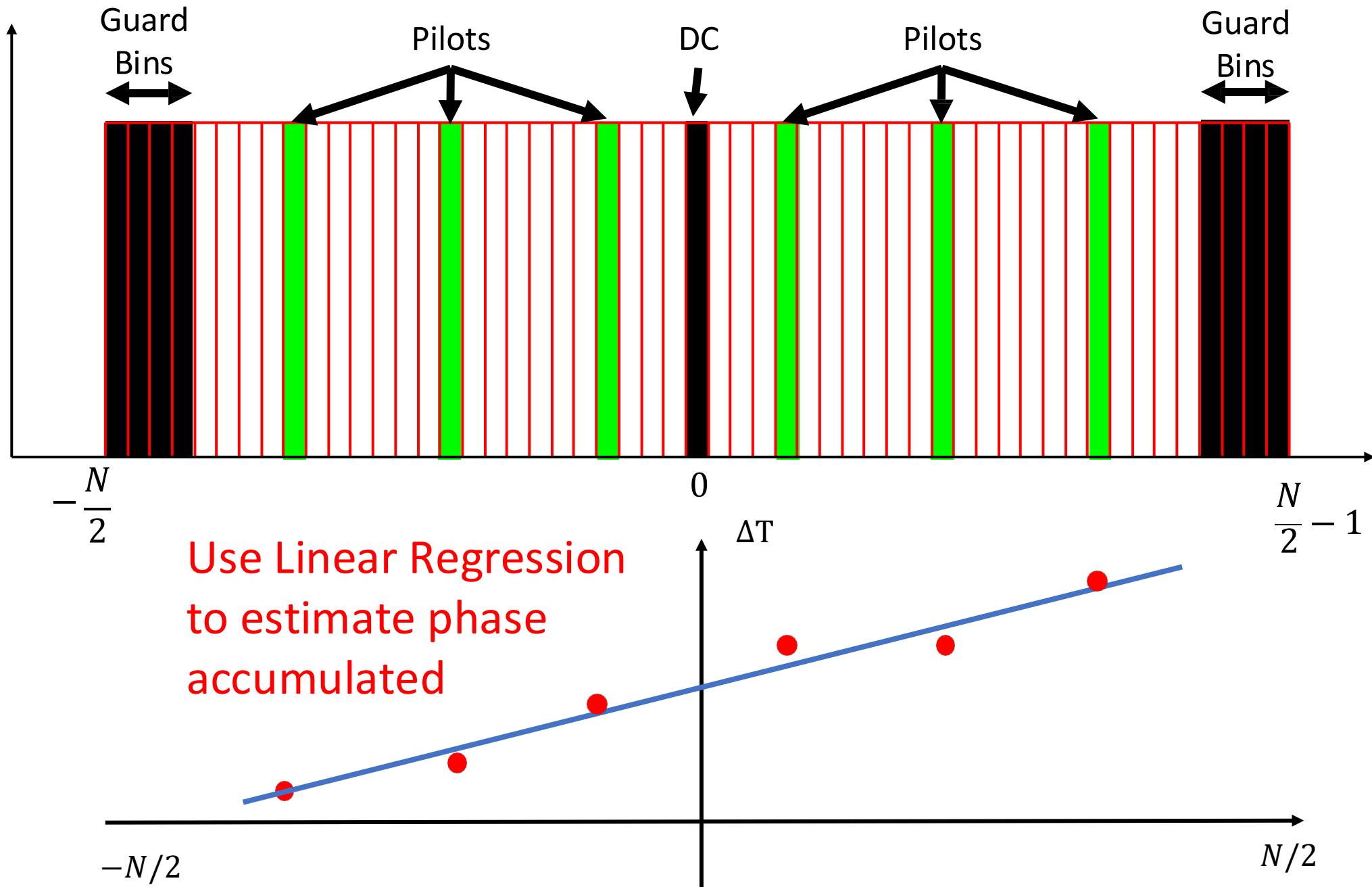
OFDM Phase Tracking

- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



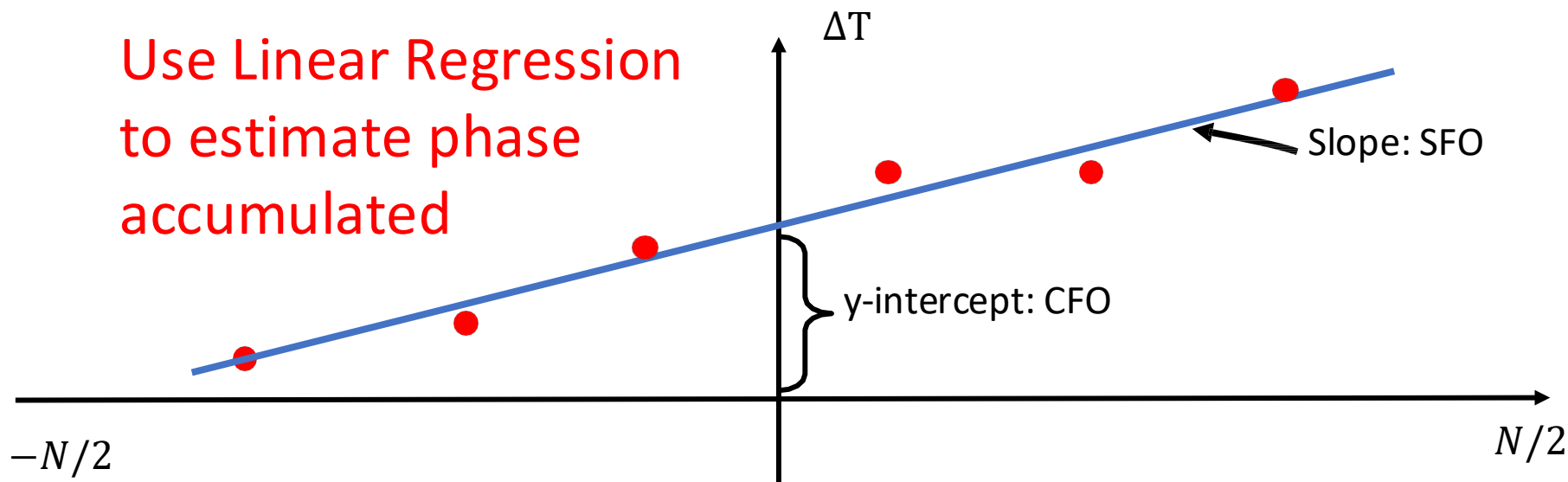
- Sufficient to estimate slope & y-intercept to know the phase accumulated for all subcarriers.
- Use only few subcarriers as pilots & send known bits in them.

OFDM Phase Tracking



OFDM Phase Tracking

Use Linear Regression
to estimate phase
accumulated

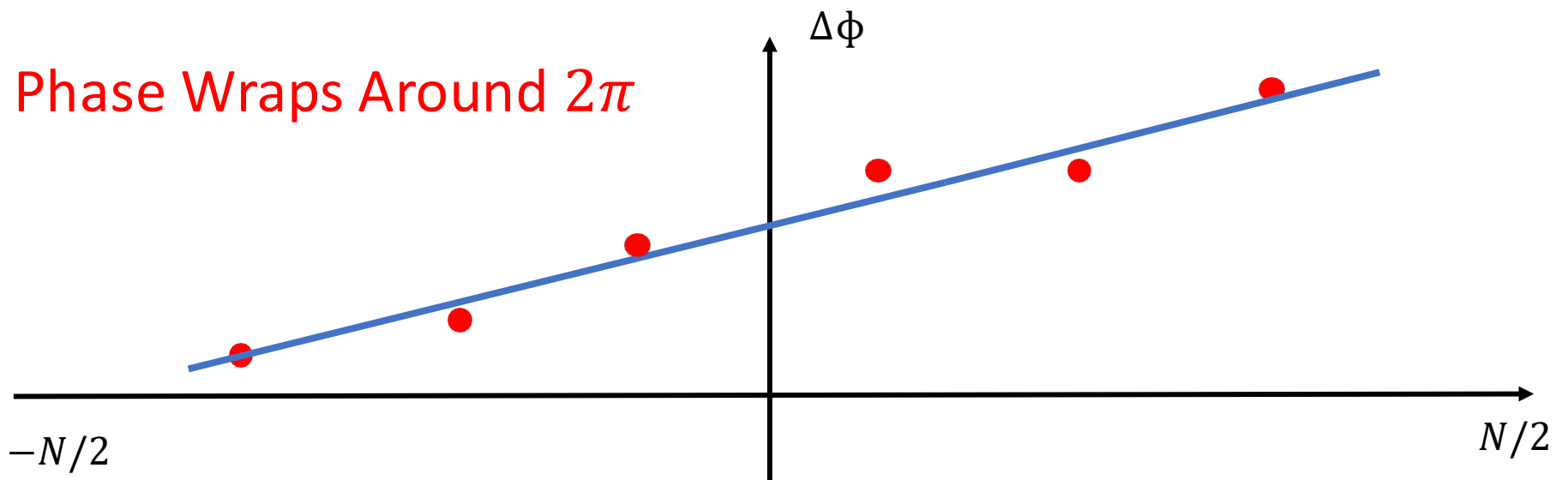


$$\text{Slope} = 2\pi \frac{(N + CP)\delta T_s}{N}$$

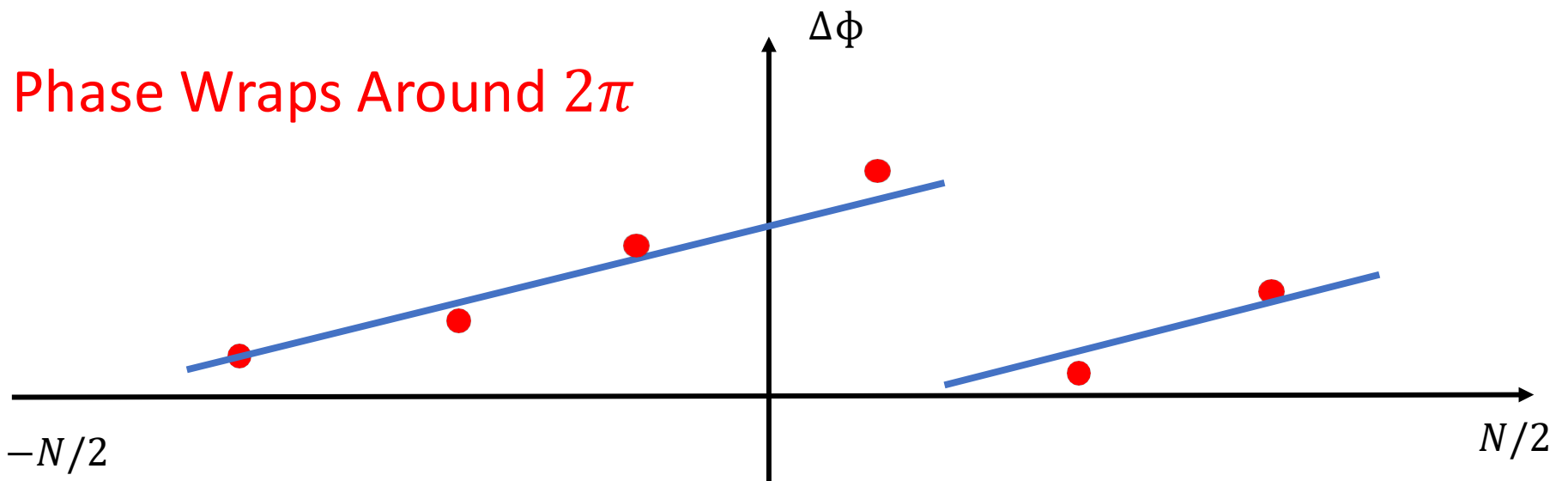
$$Y - \text{intercept} = -2\pi\delta f_c(N + CP)T_s$$

$$\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$$

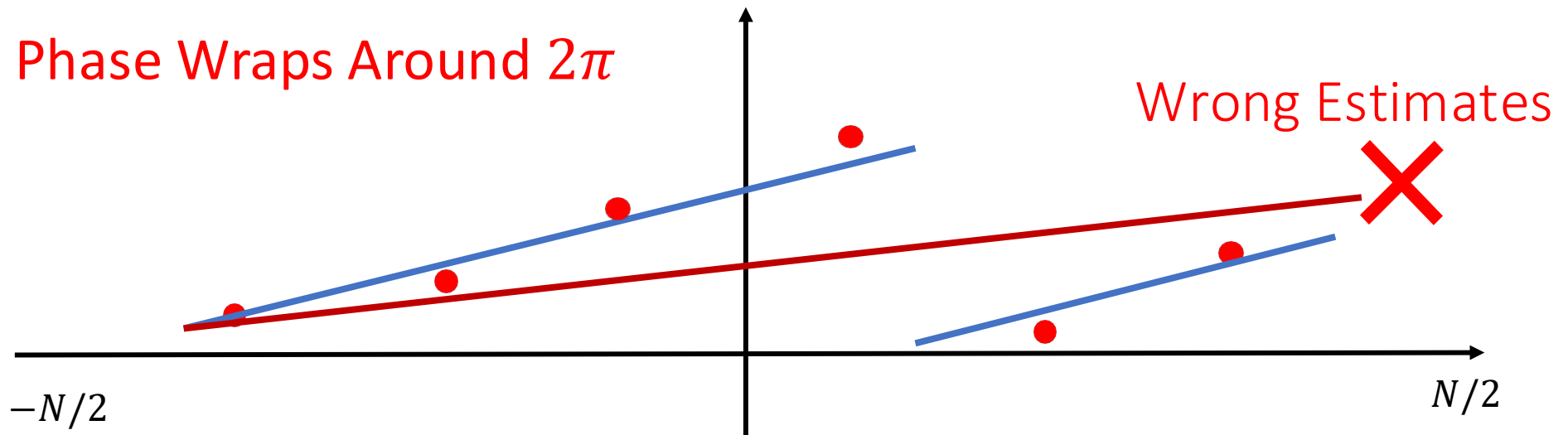
Be Careful!



Be Careful!



Be Careful!

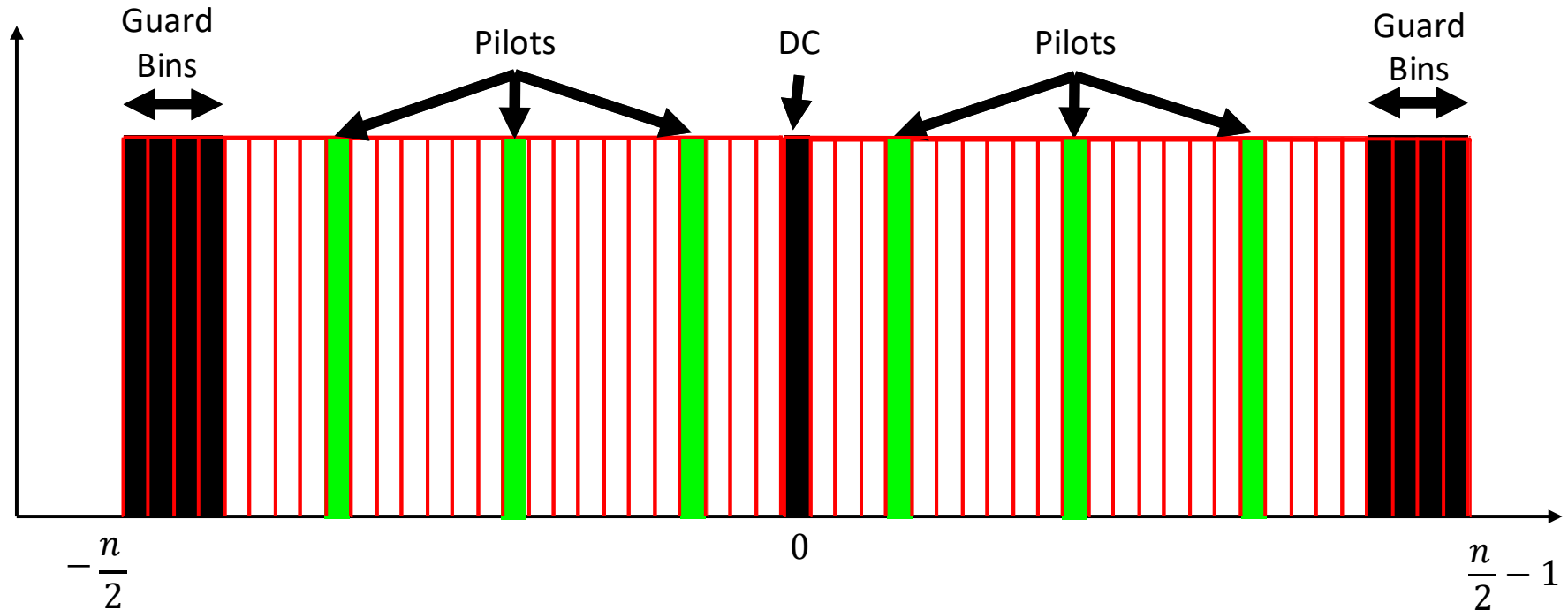


Need to unwrap phase mod 2π first! Use `unwrap()` function in MATLAB

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-n/2$ and $n/2 - 1$

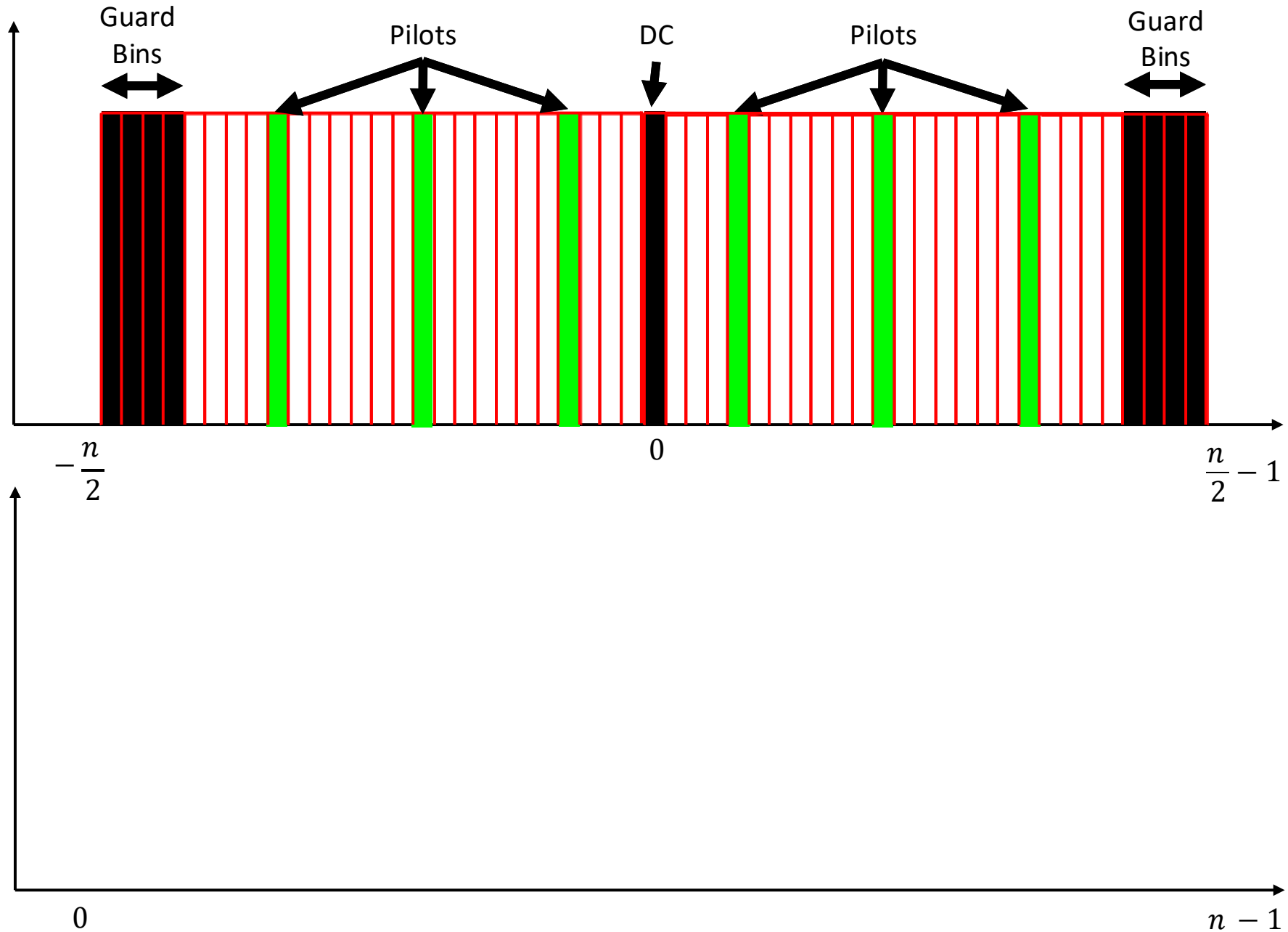
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



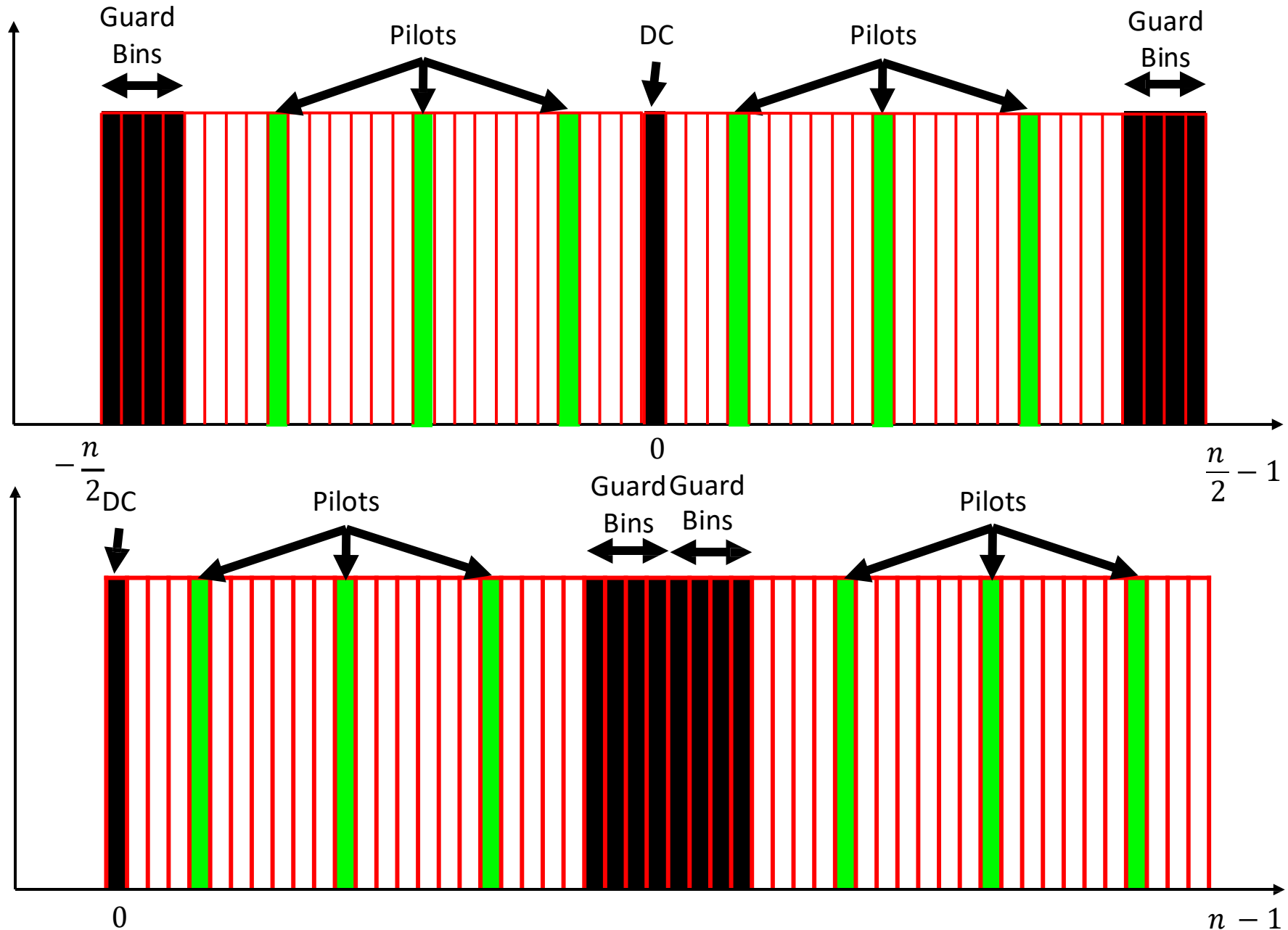
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



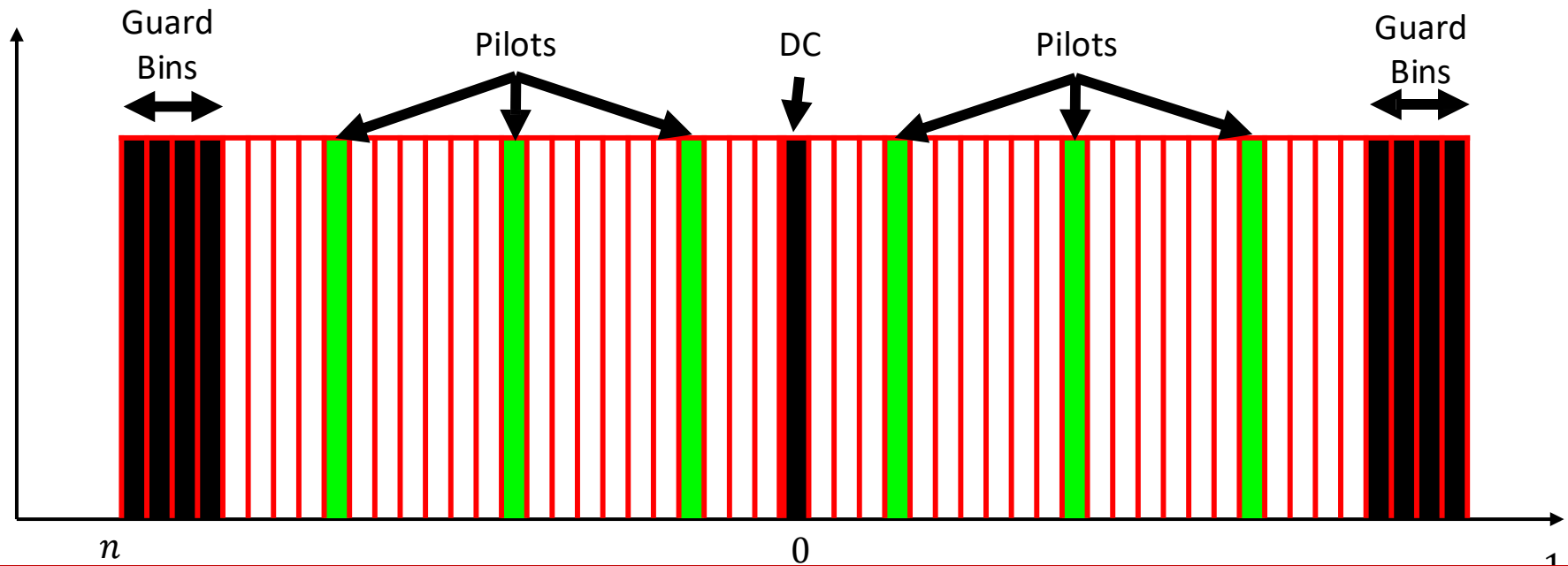
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-n/2$ and $n/2 - 1$



Need to `fftshift()` function in MATLAB + provided functions to convert indexing.

